

SYSTEM SEGMENT SPECIFICATION

Components: SEA SW OBB - Train Detection System / Eulynx
Solution Element: Virtual Interlocking
Solution: Digital Interlocking OBB
Country Austria

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Document Identifier Radical	DTC	Enterprise Identifier	Revision
	305	16282~9950	ED01 PD01
3BU 61400 0173 DTAPA, ED01 PD01			

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LOG OF CHANGES			
Revision	Date	Author	Change Description
ED01 PD01	2025-05-21	L. Roman, DEV / M. Zirwes, DEV / F. Pratser, DEV	Initial edition for OBB EEC.

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1 Scope

1.1 System Identification

Project Name:	DSTW ÖBB Entwicklung
Project ID:	100025795 / ATRELDE91
Solution Element:	ATRELDE91 VIL OBB 1.0.0
Contract:	Digitales Stellwerk DSTW - Rahmenvereinbarung
Identification:	LockTrac 6131 ELEKTRA LockTrac 6192 DCA FieldTrac 6315 AzLM
Customer:	ÖBB
End Customer:	ÖBB
SIL Level:	SIL 4

Table 1: System Identification

1.2 Identification

SEA is also called SEC in international context, and is also called EEC in product context; EEC, SEC and SEA are synonyms. This document keeps using “SEA”.

SEA consists of several components of hardware and software. All software artifacts of SEA are using the naming prefix SEA SW to distinguish them from hardware-focused documents.

The term “TDS SW” identifies the subsystem of all software within the SEA SW system, which are working together to fulfill the functionality for the Train Detection System (TDS).

The TDS SW element identifies the subsystem of all software within the SEA SW system, which are working together to fulfill the functionality for an Train Detection System controlling module.

A SEA SW system consists of

- the generic library SEA SW GEN: This implements the basic, country independent functionality including functionality common to all elements controlled by SEA, communication protocols, diagnosis and maintenance functionality, startup and shutdown mechanisms and other independent functionality common for specific SEA SW systems. SEA SW GEN is defined in the SSS document [5].

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- submodules extending the generic system with functionality specific to elements such as NIC, Point etc.: They are defined in separate documents, with Train Detection System being defined in this document.
- country-specific extensions to this library: Any ELEKTRA country version needs to adapt the desired SEA/SEC functionality offered by the generic library SEA SW GEN to the country's needs. This includes e.g. the country-specific variant of the CC/DGP - EC-MUX interface, the country-specific signalling rules, the country-specific elements.

This SSS provides the generic system requirements for the functional submodule Train Detection System, derived from high level documents - Subsystem SSS Virtual Interlocking [9], Subsystem SSDD Virtual Interlocking [10], Subsystem SSS Virtual Interlocking Security [11], Requirements Management Plan (RMP) Digital Interlocking OBB [12], SEMP Virtual Interlocking [13], Feature List Digital Interlocking OBB [14].

The architecture and design of system SEA SW GEN is part of the System Subsystem Design Description document SEA SW GEN [4], the architecture and design of the Train Detection System element is part of the Software Design Description document SEA SW Train Detection System [7].

1.3 System Overview

1.3.1 EEC: Overall System

EEC (Digital Interlocking element controlling) is a system to control trackside (outdoor) railway equipment. The main architectural approach of EEC is to locate processing units, the outdoor object controllers (OCs), next to the track to control all elements nearby. The OCs are connected to the operating ELEKTRA Virtual Interlocking Subsystem (described in Subsystem SSS Virtual Interlocking [9] and Subsystem SSDD Virtual Interlocking [10]), the element controller multiplexer (EC-MUX), through an IP based data connection over a LAN or WAN network.

The EEC implementation is done according to the Eulynx standard reference architecture [15]. To connect the indoor EC-MUX and outdoor OC components the standardized SW protocol set of Eulynx is used [15]. The OCs represent the Eulynx subsystems of type outdoor elements. Because the Eulynx reference architecture defines a separation of command and control interface (SCI, standardized command interface, implemented in SIL4 context) from diagnosis and maintenance (SDI, standardized diagnosis interface, and SMI, standardized maintenance interface) and from SSI (standardized security interface), just the SCI protocol is of interest for the EC-MUX.

EEC EC-MUX is intended to be executed in an virtualized environment within a data center with possibly high distance to the OCs / field elements. All connectivity to other components is therefore IP based.

EEC is not a stand-alone product, it is designed to be integrated into a superior control logic, which determines commands for the outdoor elements and evaluates state information sent back. EEC is able to receive commands, forward them to the outdoor elements and send back state and diagnostic information.

The EC-MUX component implements the interfaces for all types of Eulynx subsystems (object controller) for control of peripheral elements. An object controller is of a specific type according to the Eulynx subsystems:

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- OC-P - Point subsystem
- OC-LS - Light signaling subsystem
- OC-TDS - Train detection subsystem
- OC-LC - LevelCrossing subsystem
- OC-IO - Generic IO subsystem

Each system can have one or more EC-MUXes controlling several OCs:

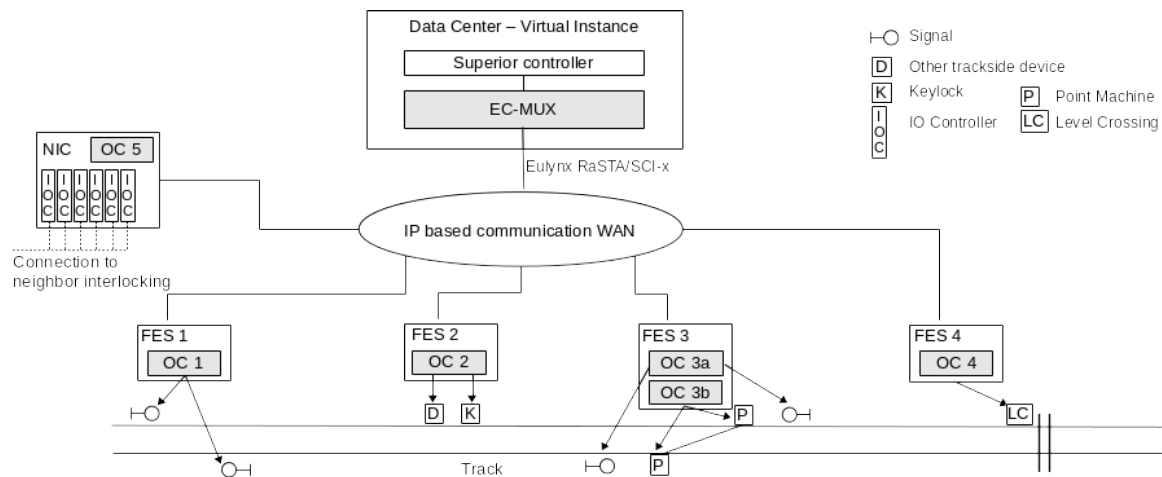


Figure 1: EEC system

- EC-MUX is the element multiplexer. It is an indoor component located in the control center and able to handle many processing units (OCs) along the track. It has the task to provide an interface for receiving commands and sending state and failure information back to the superior controlling instance.
- OCs are the object controllers inside the FES (field element station, the cabinet). It is located next to the track operating the elements nearby. The OCs represent Eulynx outdoor subsystems and are connected to the EC-MUX via Eulynx SCI protocols.

1.3.2 EEC SW GEN: EEC generic software libraries

The software of an EEC system always consists of some generic parts (EEC SW GEN libraries), which are common for all country specific implementations, and a country specific part extending the generic base libraries. Within this document, the generic parts of EEC software components are in focus. There are no site or country specific issues considered.

EEC SW GEN is the generic software library, which defines a basic system to operate OCs by implementing generic versions of the indoor component EC-MUX and implementing interfaces to the OCs.

This basic system provides all country independent functionality for EEC SW specific solutions:

- common functionality
- base communication

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- diagnosis system
- startup and shutdown mechanisms
- build mechanisms
- virtual machine support
- safety concept

EEC SW GEN is deployed as some Java software libraries, which are supposed to be included in country specific EEC SW implementations. It is assumed, that the generic libraries are always embedded into a country specific solution adding specific functionality.

EEC SW GEN implements the following generic software modules and supports the following interfaces:

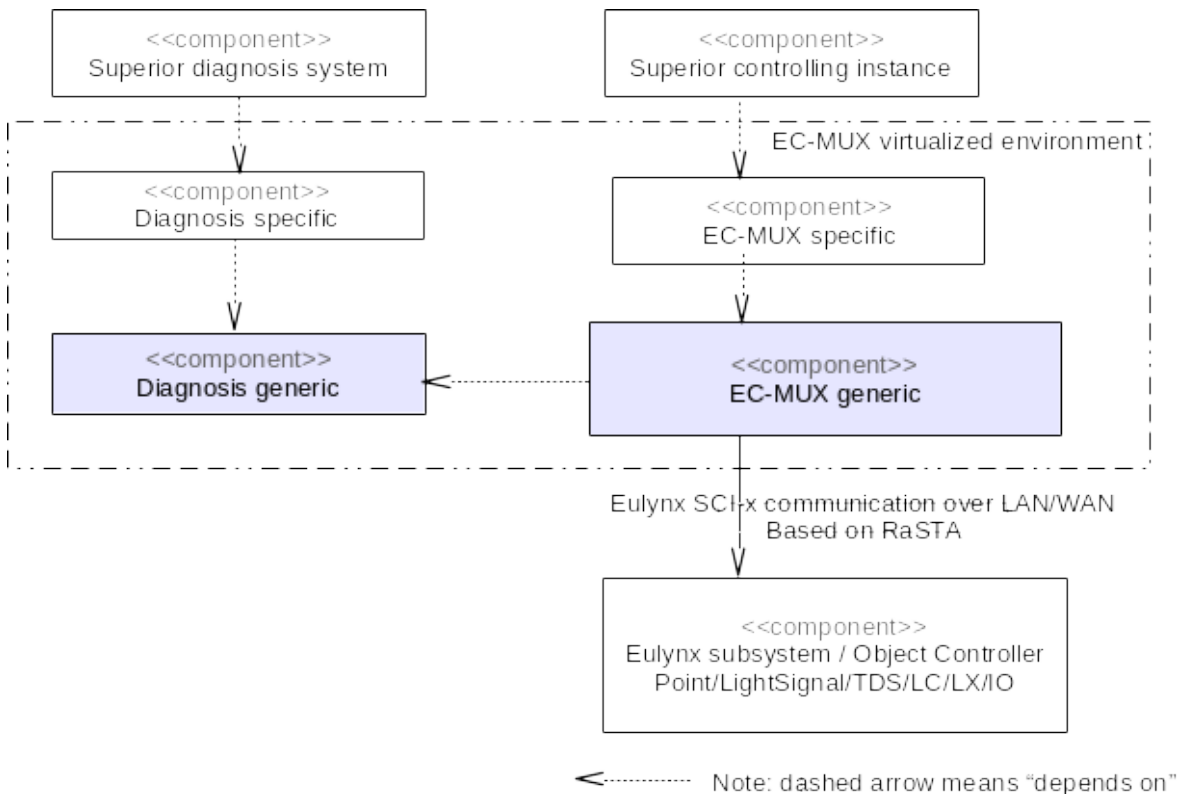


Figure 2: EEC SW GEN: main software modules and interfaces

- EEC command interface: the EC-MUX provides a country independent software interface to allow any controller using the EEC SW system to operate trackside elements and receive state and failure information
- Interface Eulynx SCI-x - IP based communication over WAN/LAN / single-message-safe principle using RaSTA: the protocol SCI-x is the Eulynx application layer interface for communication between EC-MUX and OCs. On transportation layer, according to Eulynx it is sent via the RaSTA [18] protocol
- Module EC-MUX generic: software channels for the element controller multiplexer - operates multiple OCs, provides generic functionality to send/receive command/state information to/from the correct OCs

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- Module Diagnosis generic: common instance to collect all generated diagnosis data from the whole EEC SW system

The components supposed to extend EEC SW GEN in any country specific implementation are:

- EC-MUX specific: implementation of the EEC command interface, translation of incoming commands from the superior controller to EEC commands, vice-versa for element states and failures sent back
- Diagnosis specific modules: implementation of specific translation of messages to/from a superior diagnosis system

The element specific diagnosis strategy and design are in detail described in the Software Design Description document SEA SW Train Detection System [7].

There are two types of diagnosis:

- structured diagnosis mechanisms as defined in System Subsystem Design Description document SEA SW GEN [4]
- Element Aspect Diagnosis
- Rejected Command Diagnosis
- plain text diagnosis specific for element behavior described in Software Design Description document SEA SW Train Detection System [7] and in sub-chapter *Requirement Language*.

1.3.3 Train Detection System Overview

Train Detection System (TDS) is the object controller as defined in Eulynx Requirements specification for subsystem TDS [16]. This object controller can control / supervise following HW:

- TDS for basic axle counter functionality (called Basic TDS AC)
- TDS for Train Detection Point functionality (called Basic TDS TDP)
- TDS for track circuit functionality (called Basic TDS TC)

Implementation of Axle Counter (TDS AC) functionality in SEA is described and defined in detail in sub-chapter *Axle Counter*. The TDS TDP and TDS TC are currently not supported.

Communication between SEA and HW is done via SCI-TDS:

- commands received from ELEKTRA CC are translated into SCI-TDS commands and forwarded to TDS
- SCI-TDS messages received from TDS are translated into CC states reported to ELEKTRA CC

There is no direct communication between ELEKTRA and HW. Only the TDS communicates with the HW.

PDI-version

The PDI-version of the SCI-TDS as described in EULYNX Interface specification SCI-TDS [17] is 0x04.

Protocol type

The protocol type used for the SCI-TDS is 0x20.

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1.4 Document Overview

This document is the System Subsystem Specification for the Train Detection System SW subsystem of the SEA GEN software library.

Chapter 1 defines the scope of the document, identifies the system and explains SEA and the Train Detection System subsystem.

Chapter 2 lists all references, terms, abbreviations and requirement language.

Chapter 3 of this document lists the requirements for the generic Train Detection System functionality:

- Subchapter "Operational" contains all the requirements for operational scenarios (all commands received in all states, spontaneous notifications and notifications as command answer)
- Subchapter "Technical" contains all the requirements for technical scenarios such as booting, connection loss and diagnosis.

At the end there is an index of all requirements listed in chapter Requirements.

As part of the SEA/SEC System, the Train Detection System SW subsystem fulfills the requirements from SEA SW GEN [5]. This document lists only the requirements specific to the Train Detection System.

For all functionality that is not explicitly specified, especially:

- Maintenance
- Download
- Configuration
- Diagnosis
- Error handling

generic mechanisms of SEA SW GEN are used.

1.5 Intended Audience

The target audience for this document is:

- Product Design Authority (PDA)
- Solution Engineering Manager (SEM)
- Safety Assurance Manager (SAM)
- Verifier (VER)
- Validator (VAL)
- Integration Verification, Validation, Qualification Manager (IVVQM)
- Architect (ARC)
- Requirements Manager (RM)
- Product Line Manager (PLM)

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- Quality Assurance Manager (QAM)
- Security Assurance Manager (SecAM)
- Project Security Manager (PSM)
- RAM Assurance Manager (RAM-AM)

1.6 Evolution of the document

The document is written iteratively according to the Agile Process defined in the SEA Team Charter [1]. This means that several document editions can be released before the functionality is completely described/defined. For the details on what is included in which edition of the document please see the section History below.

After the final edition containing all information for the planned functionality is released, it will be updated accordingly in case of requests for changes of the system. For type and reasons of changes please refer to the section History below.

1.6.1 History

1.6.1.1 Edition 01 PD 01

This document edition contains requirements that define the features of the TDS Axle Counter.

1.6.2 Open issues

- Axle Counter functionality as a result of reverse engineering and based on EULYNX Interface specification SCI-TDS [17] and SRS SEA OBB [6] is covered by the current edition of this document. Still, as no high-level documents - feature list, SEMP - are available it cannot be guaranteed that everything needed is covered - Subsystem SSS Virtual Interlocking [9], Subsystem SSDD Virtual Interlocking [10], Requirements Management Plan (RMP) Digital Interlocking OBB [12], SEMP Virtual Interlocking [13], Feature List Digital Interlocking OBB [14]. Editions of the referenced documents in Chapter 2. *References and Terms* need to be updated.
- Consolidate term usage SEA/SEC/EEC in the document, currently Chapter 1.2 *System Overview* uses EEC while the rest of the document uses SEA
- Add a reference to finally used versions of DCA3.x (and DCA2.x) to *System Overview*
- Traceability of system layer requirements (due to its non-availability) or to Eulynx specifications is not included in the document for now
- Security aspects to be considered in later phases of the project.
- The IDD CC/DGP - EC-MUX to be updated in ELEKTRA 4 context.
- The chapter EEC: Overall System to be updated in future.

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1.6.3 Distribution

A new version (proposal) of this document has to be distributed to all functions listed in the section Intended Audience. A new edition of this document has to be archived (online archive and conventional archive). Due to possible large numbers of involved parties the online archival of proposals is recommended. A copy for information can be downloaded from online archive by everyone in the project, who has read privilege.

1.6.4 Configuration Management

This document is generated by the Doc Maven Plugin. It is composed of textual building blocks; the content structure and metadata are defined in a "document definition" XML file. These source files are archived in the Git repository of Hitachi Rail GTS Austria, where all changes are tracked. The generated document is published on the project's website (SEA Project Site) as XHTML and PDF. Formal review proposals and released versions are archived in the Hitachi Rail GTS Austria central document repository in PDF format.

Git URL: <https://bitbucket.de.gts/scm/sea/sea>

Git commit: 4e3f2d03d4

Document definition file: /doc-obb/src/doc-definition/sss_tds.xml

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2 References and Terms

2.1 SEA Generic Documents

[1]	Team Charter SEA, 3BU 61400 0137 QMAPA, Ed. 03
[2]	Verification Cross Reference Matrix OBB-Pilot, 3BU 62300 0004 DHAPA
[3]	HITACHI-GLOSSARY, 3BU 15000 8008 WDAPQ, Ed. latest, Hitachi Rail Austria - Glossary

2.2 SEA Design Documents

[4]	SEA SW GEN SSDD - System Subsystem Design Description, 3BU 61400 0119 DSAPA, Ed. 23 - TODO: to be updated to ELEKTRA 4 architecture
-----	--

2.3 SEA Specification Documents

[5]	SEA SW GEN SSS - System Subsystem Specification, 3BU 61400 0119 DTAPA, Ed. 20 - TODO: to be updated to ELEKTRA 4 requirements
[6]	System Requirement Specification, SEA OBB, 3BU 61400 0141 DTAPA, Ed. 06 - TODO: To be extended for TDS
[7]	SEA SW OBB Train Detection System / Eulynx, Software Design Description, 3BU 61400 0173 PEAPA, Ed. 01 (planned)

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2.4 Elektra Documents

[8]	IDD CC/DGP - EC-MUX, 3BU 15000 9130 PBAPA, Ed. 07 - TODO: to be updated to ELEKTRA 4
[9]	Subsystem SSS Virtual Interlocking, 3BU 15000 6955 DTAPA, Ed. 01 (planned)
[10]	Subsystem SSDD Virtual Interlocking, 3BU 15000 6955 DSAPA, Ed. 01 (planned)
[11]	Subsystem SSS Virtual Interlocking Security, 3BU 15000 6980 DTAPA, Ed. 01 (planned)
[12]	Requirements Management Plan (RMP) Digital Interlocking OBB, 3BU 17300 1207 DUAPA, Ed. 01 PD 03
[13]	SEMP Virtual Interlocking, 3BU 15000 6955 DPAPA, Ed. 01 (planned)
[14]	Feature List - GENERIC APPLICATION - DIGITAL INTERLOCKING OBB, 3BU 17300 0000 FLAPC, Ed. 01 (planned)

2.5 External documents

[15]	[EULYNX System Definition], EULYNX System Definition, Eu.Doc.7, Baseline 4.2 (0.A)
[16]	[EULYNX Requirements specification for subsystem TDS], EULYNX Requirements specification for subsystem TDS, Eu.Doc.43, Baseline 4.2 (0.A)
[17]	[EULYNX Interface specification for SCI-TDS], EULYNX Interface specification for SCI-TDS, Eu.Doc.44, Baseline 4.1 (0.A)

2.6 Protocols

[18]	RaSTA, Rail Safe Transport Application, V03.03, DIN VDE V 0831-200, 3BU 01500 0006 PBAPA, Ed. 01
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2.7 Structure of Requirements Definition

This subsection defines the structure of one single requirement. The keywords and their explanations are shown below:

- **Identifier:** Unique identifier of the requirement
- **Type:** A classification of the different requirement types; the following keywords are defined:

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CSR Customer Specific Requirement
 DAT Data Requirement
 DOR Document Requirement
 DPR Dependability Requirement
 ECR Environmental Conservation Requirement
 FUR Functional Requirement
 GCR General Constraint Requirement
 MCR Management and Commercial Requirement
 MDR Maintenance and Diagnosis Requirement
 MIR Man-Machine Interface Requirement
 MPR Management and Production Requirement
 OAR Operational Aliveness Requirement
 PER Physical/Environmental Requirement
 PHR Prevention of Hazards Requirement
 PRR Performance and Real-Time Requirement
 PSR Product Startup Requirement
 RSR Resource Support and Consumption Requirement
 SAR Safety Requirement
 SCR Supplier to Customer Requirement
 SER Security Requirement
 TIR Technical Interface Requirement
 TVR Testing and Validation Requirement

- **Synopsis:** A brief and significant description of the requirement.
- **Description:** Complete description of the requirement. Includes, when needed, layouts (screen layouts in principle - no screen dumps), formats, and protocols for interfaces.
- **Satisfies (Origin):** Source of the requirement (feature list, reference to other requirements or internal). It is a precise backward reference of this requirement to its origin. Per convention, references to the Feature List SEA are marked FR:SEA:xxx.
- **Status:** The current status of the requirement, with the following keywords defined:
 - in work: new requirement which is not yet described completely
 - draft: requirement is written down and ready to be reviewed, it is not yet approved
 - approved: requirement was reviewed by all affected groups and is confirmed
 - descoped: Requirement delayed to a later release
 - to be descoped: requirement is planned to be descoped, no formal approval for changing the state yet
 - to be deleted: Requirement is planned to be deleted. No formal approval for deleting by CCB yet
 - deleted: Requirement is no longer valid and will not be implemented at all. Has to be agreed by CCB
- **Stability:**
 - stable: will probably not change during the life time of the product
 - volatile: subject to change
 - instable: subject to change frequently
- **Target Release:** Scheduled for which release
- **Priority:** Relative importance of fulfilling this requirement from a functional point of view (used for incremental deliveries). The following keywords are defined:

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- **Mandatory:** Requirement required by applications. If requirement is not provided, the fact and the consequences have to be documented in the validation report. The application has to decide if the product is accepted or not in absence of this requirement.
- **Optional:** Enhancement of the product may be implemented in this or a later release. Absence does not make the product unacceptable.
- **Originator:** "Equipment Practice", "Product Management", "Quality", "RAM", "Safety", "Standards /Legal", "Technics (R&D)"
- **Allocation:** Allocates requirements to the system, subsystem, component, or product environment as described in a product breakdown structure. The appropriate keywords have to be defined by the project.
- **Verification Method:** "Analysis", "Demonstration", "Inspection/Review", "Test"
- **Verification Step:** The following verification steps are defined:

Review
 Code Reading
 Analysis
 Module Test
 Sublt Test
 Integration Test
 System Test
 Performance Test
 Qualification Test
 Acceptance Test
 Field Test
 Tool Test
 Safety Approval

- **Comment:** Any additional information if necessary. (e.g. reason for deleting a requirement)

2.8 Requirement Language

This subsection defines the language used in requirements and SUBIT test cases. Note that all actions are specific to the element for which the requirement or test case is written, e.g. by received a command it is meant the command received for the element in question, by reacting or not reacting to an input it is meant that SEC is doing an action on behalf of the element in question, without considering any other elements, unless this is explicitly mentioned.

The element state mentioned is the SEA internal element state as defined in State Machine Diagram.

General:

- "SEA shall do nothing":

SEA shall do nothing in regard to the element in question:

- the state of the element is not changed
- element specific timers are not changed
- no command is sent to the HW interface or to another internal SEA component
- no message is sent on the CC and DGP interface
- "Action within t' ":

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Action is performed within the specified time interval from the last action.

- "Action after t ":

Action is performed specified amount of time after the last action.

- "Action after t from *ReferencePoint*":

Action is performed specified amount of time after the reference point.

- "Action when reaching state *State*":

Action is performed when the SEA internal element state has changed to the specified state.

- "SEA shall set the current element state to *State*":

SEA shall set the SEA internal element state to *State*.

- "SEA shall switch to *State*":

SEA shall

- set the SEA internal element state to *State*
- command the output or check-back port(s) corresponding to the element state *State* (as defined in the mapping table)
- "the current element state is *State*":

SEA internal element state is *State*.

- "the element state has changed to *State*":

SEA internal element state changed.

- "SEA shall execute EC-MUX safety reaction *SafetyReaction*"

SEA shall execute one of the safety reactions defined in SSDD SEA GEN [4], Table "EC-MUX Safety Reactions".

- "an *Element* configured as *Type*":

The element configuration corresponds as defined for the specified type.

- "initializing the element"

At SEA start-up, one of the performed actions is to read the xml database file containing station data and setup the element repository. Each element contained in the station data shall take on an initial state according to its element type configuration.

EC-MUX booting:

- "shutting down EC-MUX"

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Depending on the environment on which the EC-MUX is running on, in the case of a physical environment this step will cause a shut down of the EC-MUX software by triggering a reboot of the hardware or a shut down of the EC-MUX in the case of a virtualized environment.

- "starting up EC-MUX"

Wait until the EC-MUX is started again (for physical environment hardware is booted up again, for virtual environment EC-MUX is started). This includes also establishing the connection between EC-MUX and HW (OEC, OC...) or simulation.

Internal communication between SEA elements:

- "changing the state of *Element*":

The element (for which the requirement is written) detects a state change of another element.

CC Interface connection and fault handling:

- "SEA shall report the *Fault* fault to CC"

Send the fault information for the element from EC-MUX to CC via the s_epa_gcx_C20B_sum_fault/s_epb_gcx_D30B_sum_fault CHILL signal.

- "SEA shall report the *Fault* fault repair to CC"

Send the fault information for the element from EC-MUX to CC via the s_epa_gcx_C20C_sum_fault_rep/s_epb_gcx_D30C_sum_fault_rep CHILL signal.

- "establishing the connection to CC"

This includes successful completion of all phases required for an operational connection, like version handshake, update_history phase and general inquiry. See [8] for details.

- "losing the connection to CC"

EC-MUX does no longer see the CCx. This may either be caused by network connectivity loss, or by CCx reboot. Implicitly, while there is no connection to the CC, no communication (receiving commands or reporting) will be possible.

Connection loss to CC is detected by at least one of the following:

- CCx does not respond to EC-MUX heartbeats
- CCx does not respond to EC-MUX reactivity supervision signals
- EC-MUX gets an "ep_is_down" event
- "CC connection state is [CONNECTED|NOT_CONNECTED]":

CC connection state CONNECTED means that the connection between EC-MUX and CC is fully operational, the connection procedure as explained under "(re)establishing the connection to CC" has been performed.

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CC connection state NOT_CONNECTED means that there is no connection between EC-MUX and CC, the CC connection was never available or the disconnection procedure as explained under "losing the connection to CC" has been performed.

- "the connection to CC has been lost"

EC-MUX lost the connection to the CC. See also "losing the connection to CC".

CC interface

- "receiving the command *Command* from CC"

SEA receives a command from CC via the defined restraint-safe s_gca_gpx_2C80_peri_out /s_gcb_gpx_3D80_peri_out CHILL signal or via the defined execution-safe s_gca_gpx_2C81_2chan_out/s_gcb_gpx_3D81_2chan_out CHILL signal and process it.

DGP Interface:

- "SEA shall report the diagnosis aspect message *Aspect* to DGP":

Send e_sea_dgp_elem_aspect_reply with the given aspect (corresponding to the element state) from SEA to DGP via DGP signal s_elx_dgp_01b2_diagnose.

- "SEA shall report the *Fault* diagnosis device fault message for the *Device* device to DGP":

Send e_sea_dgp_elem_hw_fault for the element with the given faulty device (e.g. IOC) from SEA to DGP via DGP signal s_elx_dgp_01b2_diagnose.

- "SEA shall report the *Fault* diagnosis device fault message to DGP":

Send e_sea_dgp_elem_hw_fault for the element (no device specified) from EC-MUX to DGP via DGP signal s_elx_dgp_01b2_diagnose.

- "SEA shall report the *Fault* diagnosis device fault repair message for the *Device* device to DGP":

Send e_sea_dgp_elem_hw_repair for the element with the given faulty device (e.g. IOC) from SEA to DGP via DGP signal.

- "SEA shall report the *Fault* diagnosis device fault repair message to DGP":

Send e_sea_dgp_elem_hw_repair for the element (no device specified) from EC-MUX to DGP via DGP signal s_elx_dgp_01b2_diagnose.

- "SEA shall report the rejected command diagnosis to DGP":

Send e_pc_dgp_elem_cmd_info from EC-MUX to DGP via DGP signal s_elx_dgp_01b2_diagnose.

- "When receiving an aspect query from DGP":

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refers to receiving the `e_dgp_sea_elem_aspect` for the element from the DGP. SEA responds by one or more `e_sea_dgp_elem_aspect_reply` messages for the addressed element.

Related to SubIT:

- "the element configuration is completed":

This step is intended to be used only in testing at SubIT level. It represents the moment when OIF and CC filters are created and General Inquiry is sent. It is needed for identifying the specific elements when multiple elements are part of the environment setup.

Interface EC-MUX - OC connection:

- "losing the connection between EC-MUX and OC":

This OC is a placeholder and it means the step is applicable for OC-P, OC-LS, OC-TDS, OC-IO, OC-LC, OC-IO Eulynx subsystems. Eulynx connection between the EC-MUX and OC is lost, EC-MUX sets all elements controlled by this OC to the default state and executes safe commands for all elements.

- "establishing the connection between EC-MUX and OC":

This OC is a placeholder and it means the step is applicable for OC-P, OC-LS, OC-TDS, OC-IO, OC-LC, OC-IO Eulynx subsystems. Eulynx connection between the EC-MUX and OC is established, all phases required for an operational connection like version handshake and general inquiry are successfully completed.

- "SEA shall execute EC-MUX safety reaction OC Disconnection":

This step includes disconnect the connection between EC-MUX and OC, set element state to default state and report and diagnose state change and report and diagnose the disconnect.

- "SEA shall execute EC-MUX safety reaction OC-SCI Disconnection":

This step includes disconnect the connection between SCI and OC and diagnose the disconnect.

CC Interface:

- "SEA shall report the element state `CCState` with command reply flag `ReplyFlag` to CC"

SEA sends the axle counter element state `CCState` to CC via the `s_epa_gcx_C20A_gen_elem_stat/s_epb_gcx_D30A_gen_elem_stat` CHILL signal. Besides the element state itself this signal contains also `ReplyFlag`, indicating whether the state is reported as command reply or a spontaneous state change. This is represented as mode `m_logical` (true / false) in CHILL context.

- "SEA shall send the fault notification `sum_fault` with fault type `e_if_error` to CC"

SEA sends the fault information to CC via the `s_epa_gcx_C20B_sum_fault`

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/s_epb_gcx_D30B_sum_fault CHILL signal. This indicates the fault occurrence of the fault type e_if_error.

- "SEA shall send the fault notification sum_fault_rep with fault type e_if_error to CC"

SEA sends the fault information to CC via the s_epa_gcx_C20C_sum_fault_rep /s_epb_gcx_D30C_sum_fault_rep This indicates the fault repair of the fault type e_if_error.

TDS Interface:

In the following steps the parameter values stated in capital letters are symbolic values representing numeric values defined in SCI-TDS [17].

- "SEA shall forward the FC command with *FC_mode* mode to TDS"

The reset command received from CC shall be forwarded to TDS as Command FC together with the FC mode configured for this Axle Counter.

- "SEA has forwarded the FC command with *FC_mode* mode to TDS"

The FC Command received from CC has been forwarded by SEA together with the FC mode configured for this Axle Counter.

- "SEA shall close the open FC command with *FC_P* mode"

In OCCUPIED / DISTURBED states and with open FC command with FC-P mode, when receiving a status message with Occupancy SWEEPING_TRAIN_DETECTED / WAITING_FOR_SWEEPING_TRAIN from TDS, SEA shall close the open FC command.

- "the AC element is configured to support the *FC_mode* mode of the reset command"

The Axle Counter element is configured to support the referred mode of setting the Occupancy Status of a TVPS from the status OCCUPIED or DISTURBED to VACANT.

- "receiving the message Command Rejected with Reason *RejectReason* from TDS"

The TDS reports that the previously sent FC command was rejected with message Command Rejected from an OPERATIONAL or TECHNICAL reason.

- "receiving the message TVPS FC_P Failed with Reason *FailReason* from TDS"

In case that the FC command with FC-P mode has failed, the TDS sends the TVPS FC-P Failed message with one of the following reasons: INCORRECT_COUNT, TIMEOUT_EXPIRED, BOUNDING_NOT_PERMITTED, INTENTIONALLY_DELETED, OUTGOING_AXLE_TO_EXP, PROCESS_CANCELLED.

- "receiving the message TVPS FC_P_A Failed from TDS"

In case that the FC command with FC-P-A mode has failed, the TDS sends the TVPS FC-P-A Failed message.

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- "receiving the status message from TDS"

SEA receives the message TVPS Occupancy status which contains the following message parameters: Occupancy Status, Force Ability, Filling Level, POM Status, Disturbance Status, Change Trigger. The Occupancy Status can have the values VACANT, OCCUPIED, DISTURBED, WAITING_FOR_SWEEPING_TRAIN and SWEEPING_TRAIN_DETECTED. The Force Ability can have the values ABLE or UNABLE. The Change Trigger can have the values: PASSING_DETECTED, COMMAND_FROM_EIL, COMMAND_FROM_MAINTAINER, INITIAL_SECTION_STATE, TECHNICAL_FAILURE, INTERNAL_TRIGGER, NOT_APPLICABLE. The Disturbance Status can have the values TECHNICAL, OPERATIONAL, NOT_AVAILABLE. The POM Status can have the values POWER_SUPPLY_IS_OK, POWER_SUPPLY_IS_NOK, NOT_APPLICABLE. The Filling level has a numerical value between 0 and 65534, or NOT_APPLICABLE. This step is usually followed by further step(s) stating the parameter contents in particular.

- "status message contains Occupancy Status *OccupStatus*"

The received status message contains the Occupancy Status of the TVPS which can have the values VACANT, OCCUPIED, DISTURBED, WAITING_FOR_SWEEPING_TRAIN and SWEEPING_TRAIN_DETECTED. The status message is the equivalent of SCI-TDS message TVPS Occupancy Status.

- "status message contains Force Ability *Ability*"

The received status message contains the Force Ability of the TVPS which can have the values ABLE or UNABLE. The status message is the equivalent of SCI-TDS message TVPS Occupancy Status.

- "receiving the status message from TDS within DelayToFree s"

SEA receives the message TVPS Occupancy Status from TDS before DelayToFree seconds are elapsed since receiving Occupancy Status VACANT. The DelayToFree value is applicable to each axle counter element and can have different values. The DelayToFree value is greater than 0s and less or equal than 16s. The status message is the equivalent of SCI-TDS message TVPS Occupancy Status.

- "the state FREE was previously reached by receiving a spontaneous status message from TDS"

The FREE state was reached by receiving a spontaneous message TVPS Occupancy Status with Occupancy Status VACANT and Force Ability UNABLE, i.e. the FREE state was not a result of FC command execution. The status message is the equivalent of SCI-TDS message TVPS Occupancy Status.

- "SEA shall process the message"

Incoming status message has passed the consistency check, and shall be processed according to the defined element process. The relevant message parameters are Occupancy Status and Force Ability. Particular value of every other attribute of the status message is checked for consistency only and do not influence the element state. For the state transitions / reporting / diagnosis the requirements for regular behavior are applicable. The status message is the equivalent of SCI-

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TDS message TVPS Occupancy Status.

- "receiving the status message with Occupancy Status *OccupState* and Force Ability *ForceAbility* from TDS"

The status message contains the Occupancy Status of the TVPS and the Force Ability corresponding to the given parameters. The Occupancy Status of the TVPS can have the values VACANT, OCCUPIED, DISTURBED, WAITING_FOR_SWEEPING_TRAIN and SWEEPING_TRAIN_DETECTED. The Force Ability can have the values ABLE or UNABLE. Particular value of every other attribute of the status message is irrelevant.

- "receiving the status message with Occupancy Status *OccupStatus* and Change Trigger *ChangeTrigger* from TDS"

The status message contains the Occupancy Status of the TVPS and the Change Trigger corresponding to the given parameters. The Occupancy Status of the TVPS can have the values VACANT, OCCUPIED, DISTURBED, WAITING_FOR_SWEEPING_TRAIN and SWEEPING_TRAIN_DETECTED. The Change Trigger can have the values: PASSING_DETECTED, COMMAND_FROM_EIL, COMMAND_FROM_MAINTAINER, INITIAL_SECTION_STATE, TECHNICAL_FAILURE, INTERNAL_TRIGGER, NOT_APPLICABLE. Particular value of every other attribute of the status message is irrelevant.

- "receiving the status message with Occupancy Status *OccupStatus* and Disturbance Status *DisturbanceStatus* from TDS"

The status message contains the Occupancy Status of the TVPS and the Disturbance Status corresponding to the given parameters. The Occupancy Status of the TVPS can have the values VACANT, OCCUPIED, DISTURBED, WAITING_FOR_SWEEPING_TRAIN and SWEEPING_TRAIN_DETECTED. The Disturbance Status can have the values: TECHNICAL, OPERATIONAL, NOT_AVAILABLE. Particular value of every other attribute of the status message is irrelevant.

- "receiving the status message with POM Status *POM_Status* from TDS"

The status message contains the POM Status corresponding to the given parameters. The POM Status can have the values POWER_SUPPLY_IS_OK, POWER_SUPPLY_IS_NOK and NOT_APPLICABLE. Particular value of every other attribute of the status message is irrelevant.

- "receiving the status message with Disturbance Status *DisturbanceStatus* from TDS"

The status message contains the Disturbance Status of the TVPS corresponding to the given parameters. The Disturbance Status can have the values TECHNICAL, OPERATIONAL or NOT_AVAILABLE. Particular value of every other attribute of the status message is irrelevant.

- "receiving the status message with Filling Level NOT_APPLICABLE / different from NOT_APPLICABLE from TDS"

The status message contains Filling Level corresponding to the given parameters. The Filling Level has a numerical value between 0 and 65534, or NOT_APPLICABLE. Particular value of every other attribute of the status message is irrelevant.

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- "receiving the TDP Status message with State of Passing *StateOfPassing* and Direction of Passing *DirectionOfPassing* from TDS"

The TDP Status message contains the State of Passing and the Direction of Passing corresponding to the given parameters. The State of Passing can have the values PASSED, NOT_PASSED or DISTURBED. The Direction of Passing can have the values REFERENCE_DIRECTION, AGAINST_REFERENCE_DIRECTION or WITHOUT_INDICATED_DIRECTION.

- "receiving the Additional Information message with Wheel Diameter value within valid range and Speed value within valid range from TDS"

The Additional Information message received from TDS contains the Wheel Diameter and the Speed. According to SCI-TDS the Wheel Diameter is within 0x0001 and 0x9999 and the Speed is within 0x0000 and 0x9999.

- "TDS Occupancy Status is *OccupStatus*"

In requirements, SEA Axle Counter is referenced and not the TDS directly. When the TDS is not connected, SEA does not have the information about the state of the TDS Occupancy Status and SEA element state cannot be directly reference as it is different from the real TDS Occupancy Status. Not to list states of the TDS Occupancy Status directly, this expression is used. The TDS Occupancy Status can have the values VACANT, OCCUPIED, DISTURBED, WAITING_FOR_SWEEPING_TRAIN and SWEEPING_TRAIN_DETECTED.

- "TDS Force Ability is *ForceAbility*"

In requirements, SEA Axle Counter is referenced and not the TDS directly. When the TDS is not connected, SEA does not have the information about the state of the TDS Force Ability and SEA element state cannot be directly reference as it is different from the real TDS Force Ability. Not to list states of the TDS Force Ability directly, this expression is used. The TDS Force Ability can have the values ABLE or UNABLE.

2.9 Abbreviations and Acronyms

For Terms and abbreviations see [3].

2.10 Definitions and Terms

2.10.1 Definitions

1. Configuration data
any information or file needed by the application software when executed
(station data and software configuration files)

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2. Software configuration files
files, which are loaded by the application at startup to read parameters for executing the software; not site specific
3. Station data
the site specific preparation data, which describes the present elements of a component; output from Orest in XML format
4. Application software (files)
the executables (files), which are executed when the application is started
5. Element process
The EC-MUX contains the logic corresponding to the behavior of each configured element, internally divided into a set of Java classes. This logic is referred to as "element process".
6. Initial element state
The initial state is the state of the element upon startup when DCA is disconnected.
7. Default element state
The default state is the one the element shall try to recover to when an error occurs.
8. Fault
Defect, mistake or inaccuracy which could result in failure or in a deviation from the intended performance or behavior
9. Failure
Unacceptable difference between required and observed performance.

To describe parts of code, the following terms are used:

- Class
(usually) small part of executable code, encapsulating its own data in object-oriented context.
Example: "at.thales.sea.oec.builders.LampDeviceCfg"
- Package
group of classes contextually fitting together (e.g. depending on each other). Example: "at.thales.sea.oec.builders". Grouping may be applied recursively; a package may contain sub-packages.
- Module
A Maven module. Consists of one or more packages. This is the smallest unit that can be built on its own. Example: "ecbase".
- Component
Software part that serves a defined task. Consists of one or more modules. Note that modules may be used in multiple components, so in general components share the same modules.
Example: "EC-MUX generic". Note further, that this is not the definition of the term component as stated in CENELEC standard, but a term used in SW context.
- Deliverable
Software part that may be deployed on its own. This includes binaries, scripts, and configuration items like resource files or station data.
- CC command (peri_com) to SCI-TDS command

CC command	configured FC type	Function	SCI-TDS command	Parameter
e_reset_rp	conditional	reset the AC count only if the ability to be forced to clear is given	Command "FC"	Mode of FC: FC_C
			Command	Mode of

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e_reset_rp	unconditional	reset the AC count always	"FC"	FC: FC_U
e_reset_rp	preparatory	reset the AC count considering the sweeping train before	Command "FC"	Mode of FC: FC_P

Table 2: Mapping CC command (CC to EC-MUX) to SCI-TDS command (EC-MUX to TDS)

- CC command (2chan_com) to SCI-TDS command

There is no 2chan_com command addressed to the Axle Counter element.

- SCI-TDS Message "TVPS Occupancy Status" to CC state message (gen_elem_stat)

SCI-TDS Occupancy Status	Function	CC state message
vacant	vacant / free outgoing = incoming	e_track_free_es
occupied	occupied outgoing < incoming	e_track_occupied_es
disturbed	disturbed outgoing > incoming(counting error)	e_track_not_regular_es
waiting for a sweeping train after FC-P-A or FC-P command	special value in context of pending FC-P-A / FC-P command	none
waiting for an acknowledgement after FC-P-A command	request for acknowledgement by interlocking system	none, not applicable in ELEKTRA context
sweeping train detected	special value in context of pending FC-P-A / FC-P command	none
default state after startup or after connection loss between EC-MUX and TDS, see chapter Connection handling - EC-MUX - TDS connection	undefined	e_undefined_es

Table 3: Mapping SCI-TDS Message TVPS Occupancy Status (TDS to EC-MUX), attribute Occupancy Status to CC element state message (EC-MUX to CC)

Remark: The term incoming / outgoing means the axle count of incoming / outgoing train axles. For details see chapter 3.1 Axle Counter. The attribute "Occupancy Status" is the only one out of attributes transmitted by the Message "TVPS Occupancy Status" which results in state reporting to the CC. No mappings needed for any other attribute.

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- SCI-TDS Message "TVPS Occupancy Status" to CC summary fault message (sum_fault / sum_fault_rep)

SCI-TDS Occupancy Status	Fault information	CC summary fault
vacant	not faulty	sum_fault_rep
occupied	not faulty	sum_fault_rep
disturbed	faulty	sum_fault
undefined	faulty	sum_fault
any other value in FC-P context	unchanged	none

Table 4: Mapping SCI-TDS Message TVPS Occupancy Status (TDS to EC-MUX), attribute Occupancy Status to CC fault / repair message (EC-MUX to CC)

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3 Requirements

3.1 Axle Counter

An Axle Counter element (AC) consists in the HW of two Train Detection Points (TDP) counting the incoming / outgoing train axles. Based on the comparison of incoming / outgoing axle count the TDS evaluates the current Occupancy Status of the AC: vacant (= free in ELEKTRA context), occupied, disturbed (in case of a counting error or if HW is disconnected).

Eulynx Axle Counter is in SEA realized via the component `:eulx_azlm` as shown in the following figure:

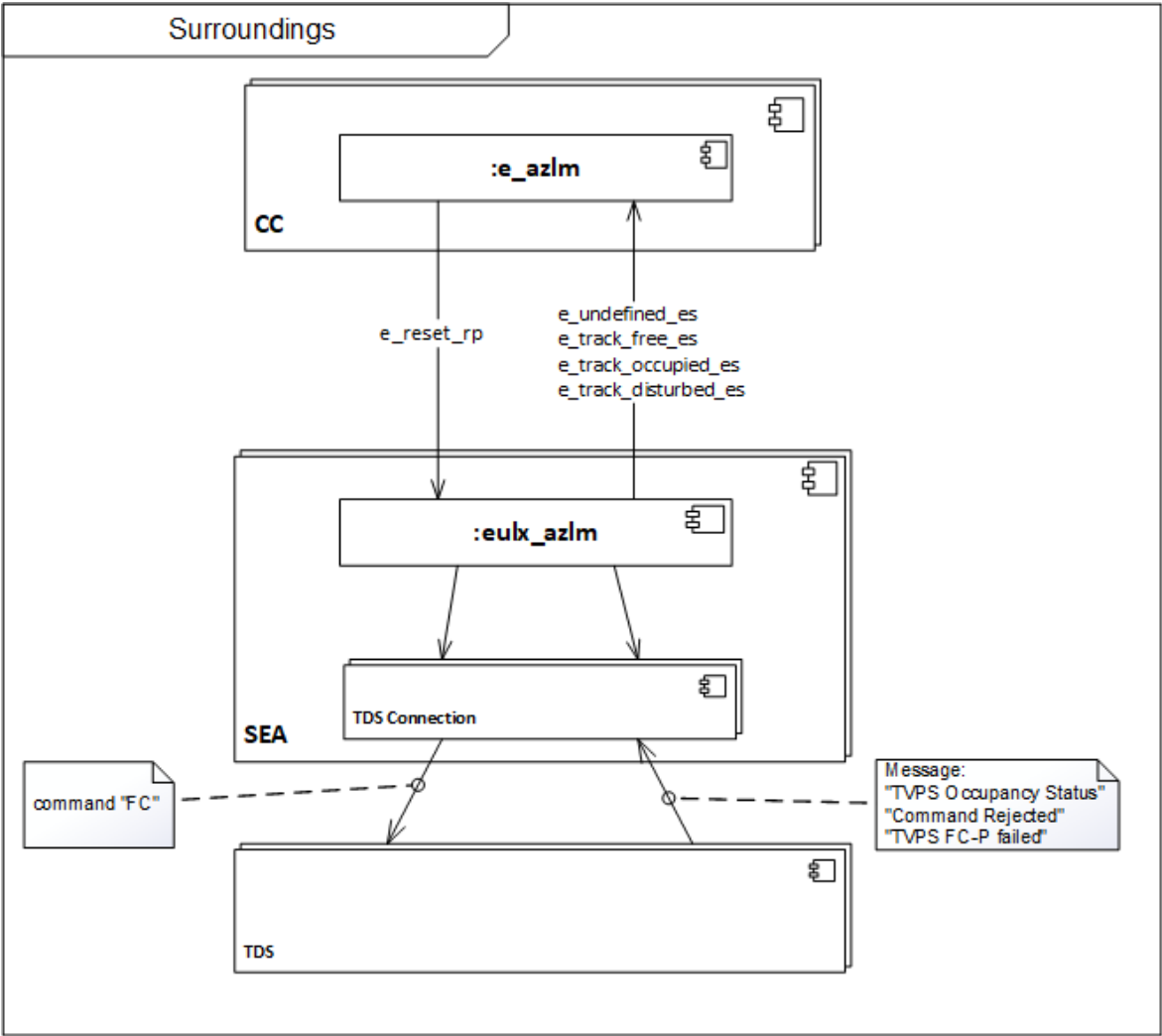


Figure 3: SEA Eulynx Axle Counter - Surroundings

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For details regarding CC IF and its relation to SCI-TDS see sub-chapter *Definitions and Terms*.

SCI-TDS Command / Message	Value	Supported by SEA / ELEKTRA
command "FC"	0x0001	yes
command "Update Filling Level"	0x0002	no
command "Cancel"	0x0008	no
command "Disable the restriction to force section status to clear"	0x0003	no
message "TVPS Occupancy Status"	0x0007	yes
message "Command Rejected"	0x0006	yes
message "TVPS FC-P failed"	0x0010	yes
message "TVPS FC-P-A failed"	0x0011	no
message "Additional Information"	0x0012	no
message "TDP Status"	0x000B	no

Table 5: Support of SCI-TDS command/messages in SEA/ELEKTRA context

State Notifications

TDS reports the current AC state to the interlocking system by message "TVPS Occupancy Status" containing several attributes. Out of them only the Occupancy Status shall be forwarded to the CC, every other attribute will be used just for internal / diagnosis purpose or not used at all:

- Occupancy Status
 - vacant / occupied / disturbed - corresponding states to be reported to the CC; for details see sub-chapter *Definitions and Terms* and the state machine diagram
 - In context of the preparatory reset execution TDS reports the Occupancy Status with special sweeping train values. These values shall never be reported to the CC, as there is no corresponding CC IF.
For further details see paragraph "TDS replies to reset command" below.
- Ability to be forced to clear
condition for the admissibility of the reset command, for internal use only
- Filling Level
current axle count
to be used only in context of the command "Update Filling Level"
this command is not implemented in the CC level and thus never be executed, this attribute shall be ignored
- POM Status
power supply status of a Track Circuit element; not applicable for AC, to be ignored
- Disturbance Status
additional information about the disturbance cause, to be provided by TDS in disturbed state, for diagnosis purpose only

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- Change Trigger
information about the reason for sending this status message, e.g. execution of the command by interlocking system / on-site, spontaneous message

Commands

In ELEKTRA context there is only one command addressed to the AC element: command reset (the axle count to 0), ger. Achszählergrundstellung, which is a documented command.

In Eulynx context this command is called "force to clear" (FC). There are several possible modes for the FC command. As it is impossible to distinguish between different FC modes when using the current CC IF, the FC mode applicable for particular AC instance shall be configurable. (Same solution implemented for the AZLM via OCS-SB in traditional EC.) Following FC modes shall be configurable:

- unconditional: reset the AC count always, i.e. despite its current ability to be forced to clear
- conditional: reset the AC count only if the ability to be forced to clear is given
- preparatory: reset the AC count considering the sweeping train before

Remark: SCI-TDS offers also the preparatory FC command with acknowledgement, which requires an additional communication with the interlocking system. As this communication is not implemented in the ELEKTRA CC level yet, this FC mode will currently not be considered in EC-MUX and never forwarded to TDS.

Command admissibility

SEA itself evaluates first the admissibility of the reset command received from the CC, based on the current

- occupation status:
reset command to be forwarded to TDS only if the current occupation state is not free
- ability to be forced to clear:
conditional reset is not allowed if the ability to be forced to clear is currently not given;
if the AC is currently free, then its always unable to be forced to clear
- configured FC mode:
different conditions regarding the ability to be forced to clear for different FC modes (see above)

If the reset command is admissible, then it shall be forwarded to TDS by Command "FC", applying the FC mode corresponding to the AC configuration.

TDS replies to reset command

Conditional / unconditional reset:

- successful execution
Message "TVPS Occupancy status" with
Occupancy Status: vacant and
Ability to be forced to clear: unable to be forced to clear
- rejection
TDS replies by the message "Command Rejected"

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Preparatory reset:

In the context of the preparatory reset there is a frequently used term “sweeping train”. Especially in the OMC the operator is not present in particular station, where the problem occurs, so it is impossible to check the situation by sight (maybe there is really a single carriage / some visible obstacle causing this problem). In this situation the operator can give an order to let a special train = sweeping train pass this area, with a very slow speed (ger. auf Sicht fahren). If no obstacle is detected and the train is able to pass, then

- the counting is maybe cleared by the sweeping train itself and
- it is really possible to execute the reset operation

TDS replies to the preparatory reset command

- successful execution\ sequence of messages “TVPS Occupancy status” with Occupancy Status:
 1. waiting for a sweeping train after FC-P-A or FC-P command
 2. sweeping train detected
 3. vacant

Each of them associated with unable to be forced to clear. * rejection

Before the message sequence described above for the successful execution is completed, TDS replies by the message 3.4.7 Message “TVPS FC-P failed”.

The preparatory reset with acknowledgement is not going to be supported by SEA, therefore 3.4.8 Message “TVPS FC-P-A failed” is not applicable.

For further details see [17] chapter

- 3.4.1 Command “FC”
- 3.4.5 Message “TVPS Occupancy Status”
- 3.4.6 Message “Command Rejected”
- 3.4.7 Message “TVPS FC-P failed”

and command scenarios defined in [16].

CC always needs a reply for a command forwarded to EC / EC-MUX. Before receiving the command reply CC will send no further command addressed to the same AC instance. For this reason SEA needs to reply each rejected reset command by repeating the current occupation state, with command reply flag set to true. This is applicable for:

- rejection by EC-MUX itself, without forwarding the command to TDS
- rejection by TDS, by message “Command Rejected” / “TVPS FC-P failed”

Connection handling

EC-MUX - TDS connection

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When receiving an unsupported message or a supported message with inconsistent parameters the safety reaction "OC Disconnection" will be performed. By this means not only the Axle Counter element which the causing message was addressed to will be inoperable till re-establishing the connection between EC-MUX and TDS, but also every other element (AC, TDP) controlled by the same OC, i.e. same TDS sub-system.

Due to the connection loss between EC-MUX and TDS the current AC element state shall be set to the default state UNDEFINED and reported to CC. If the previous state was not faulty (FREE, OCCUPIED_ABLE, OCCUPIED_UNABLE, OCCUPIED_TIMED), then the fault occurrence shall be reported to CC as well.

When (re-)establishing the connection between EC-MUX and TDS state transition shall be triggered based on the state message "TVPS Occupancy Status" received from TDS during the PDI connection establishment (transmission of the status information):

- for Occupancy State VACANT transition to FREE
- for Occupancy State OCCUPIED with Ability to be forced to clear "TVPS is able to be forced to clear" transition to OCCUPIED_ABLE
- for Occupancy State OCCUPIED with Ability to be forced to clear "TVPS is not able to be forced to clear" transition to OCCUPIED_UNABLE
- for Occupancy State DISTURBED with Ability to be forced to clear "TVPS is able to be forced to clear" transition to DISTURBED_ABLE
- for Occupancy State DISTURBED with Ability to be forced to clear "TVPS is not able to be forced to clear" transition to DISTURBED_UNABLE
- for Occupancy State with sweeping train values, stay in UNDEFINED as these values should not be generated by the TDS without the FC-P command

CC connection

No safe command shall be executed due to the CC connection loss, i.e. AC will keep its current element state. Updates of the element state provided by TDS while the CC connection is lost shall be accepted.

When (re-)establishing the connection to CC, the current element state shall be reported to CC. If the current element state is not faulty (FREE, OCCUPIED_ABLE, OCCUPIED_UNABLE, OCCUPIED_TIMED), then the fault repair shall be reported to CC. If the current element state is faulty (UNDEFINED, DISTURBED_ABLE, DISTURBED_UNABLE, DISTURBED_TIMED), then the fault occurrence shall be reported to CC.

Element state machine

Following internal element states are defined for the Axle Counter:

Axle Counter State	Functionality	state reporting to CC
UNDEFINED	initial state / default state after TDS connection loss	e_undefined_es
FREE	vacant state notified by TDS as reply to FC command or\ in case of spontaneous msg reporting timer expired already	e_track_free_es
	occupied state + able to be forced to clear notified	

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OCCUPIED_ABLE	by TDS	e_track_occupied_es *)
OCCUPIED_UNABLE	occupied state + unable to be forced to clear notified by TDS	e_track_occupied_es *)
OCCUPIED_TIMED	vacant state notified by TDS as spontaneous msg after previous occupied state\ reporting timer not expired yet	e_track_occupied_es **)
DISTURBED_ABLE	disturbed state + able to be forced to clear notified by TDS	e_track_not_regular_es *)
DISTURBED_UNABLE	disturbed state + unable to be forced to clear notified by TDS	e_track_not_regular_es *)
DISTURBED_TIMED	vacant state notified by TDS as spontaneous msg after previous disturbed state\ reporting timer not expired yet	e_track_not_regular_es **)
CHANGING_TO_FREE	the current element state is one of OCCUPIED_ABLE, OCCUPIED_UNABLE, DISTURBED_ABLE, DISTURBED_UNABLE and the element has an open FC command	Not Applicable

Table 6: Axle Counter - Internal Element States

Remark:

) e_track_occupied_es / e_track_not_regular_es to be reported only if the state has been reached from another state than OCCUPIED_ / DISTURBED_*

**) never to be reported at the transition to this state, only at CC re-connect

SEA would accept each status message received from TDS and update the current element state according to the Occupancy Status and Ability to be forced to clear contained in the message.\ It is possible that some transitions will never happen in regular operation.

In [16] there are some scenarios with both attributes changed, in particular the transitions from OCCUPIED_ABLE to DISTURBED_UNABLE and from OCCUPIED_UNABLE to DISTURBED_ABLE, while there are no explicit scenarios for the transitions from DISTURBED_ABLE to OCCUPIED_UNABLE and DISTURBED_UNABLE to OCCUPIED_ABLE. Nevertheless these transitions will be accepted as well as SEA cannot reasonably refuse a state reported by TDS.

The state machine of the Axle Counter element is shown in the following figure:

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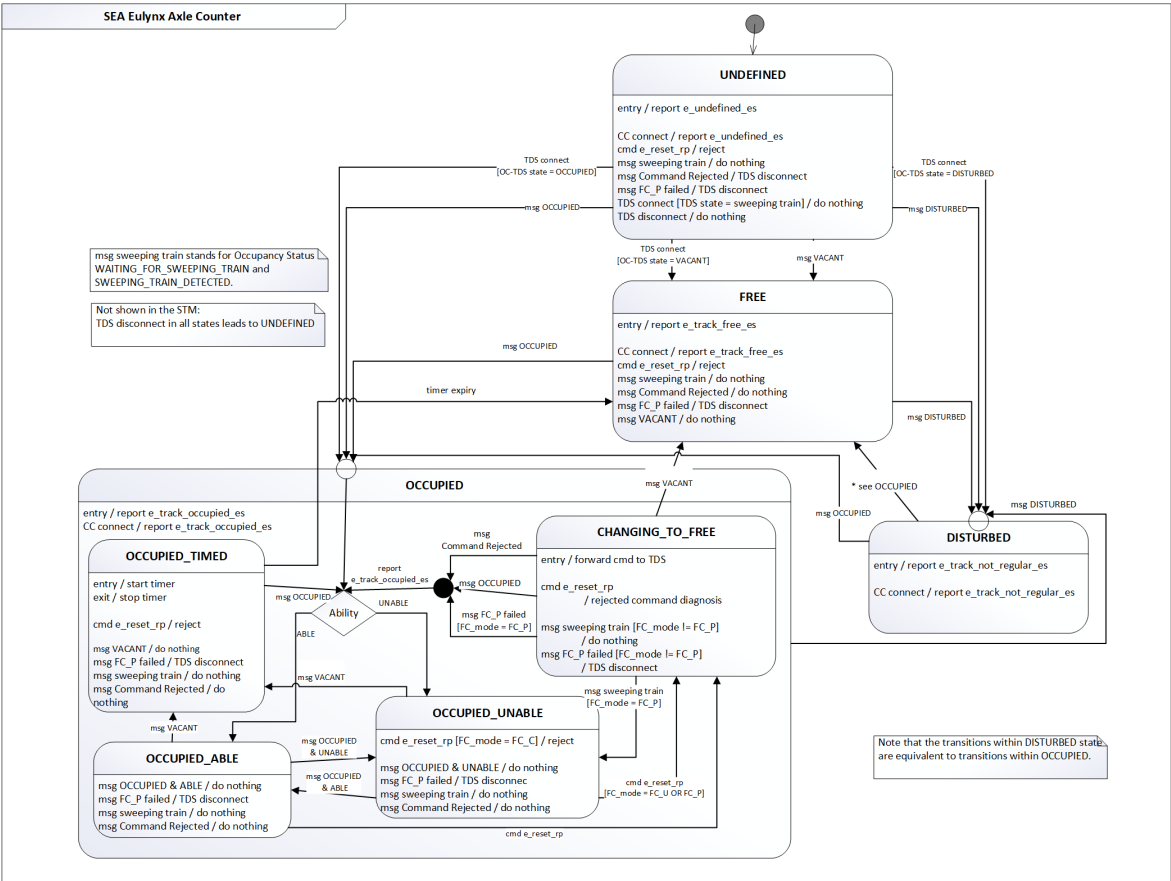


Figure 4: SEA Eulynx Axle Counter - State Machine

3.1.1 Operational - Axle Counter

3.1.1.1 Operational - Axle Counter - Command

In order to control the element state of the Axle Counter
As a SEA
I want to be able to receive the reset command from CC and and forward it to the TDS considering the configured FC mode.

For requirement language see Chapter 2.6 "Requirement Language".

Given SEA with a configured Axle Counter element
And the mapping between SEA diagnosis aspect and DGP diagnosis value is defined

SEA diagnosis aspect	DGP diagnosis value
AC_ FREE	0

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SEA diagnosis aspect	DGP diagnosis value
AC_OCCUPIED_ABLE	1
AC_OCCUPIED_UNABLE	2
AC_OCCUPIED_TIMED	3
AC_DISTURBED_ABLE	4
AC_DISTURBED_UNABLE	5
AC_DISTURBED_TIMED	6
AC_UNDEFINED	7

EC-MUX-AC:SSS:REQ: 2130	Axle Counter - Execute reset command																						
Given the current element state is <State> And the Axle Counter element is configured to support the <FC_mode> mode of the reset command When receiving the command e_reset_rp from CC Then SEA shall forward the FC command with <FC_mode> mode to TDS <table> <tr> <th>State</th><th>FC_mode</th></tr> <tr> <td>OCCUPIED_ABLE</td><td>FC_C</td></tr> <tr> <td>OCCUPIED_ABLE</td><td>FC_U</td></tr> <tr> <td>OCCUPIED_ABLE</td><td>FC_P</td></tr> <tr> <td>DISTURBED_ABLE</td><td>FC_C</td></tr> <tr> <td>DISTURBED_ABLE</td><td>FC_U</td></tr> <tr> <td>DISTURBED_ABLE</td><td>FC_P</td></tr> <tr> <td>OCCUPIED_UNABLE</td><td>FC_U</td></tr> <tr> <td>OCCUPIED_UNABLE</td><td>FC_P</td></tr> <tr> <td>DISTURBED_UNABLE</td><td>FC_U</td></tr> <tr> <td>DISTURBED_UNABLE</td><td>FC_P</td></tr> </table>		State	FC_mode	OCCUPIED_ABLE	FC_C	OCCUPIED_ABLE	FC_U	OCCUPIED_ABLE	FC_P	DISTURBED_ABLE	FC_C	DISTURBED_ABLE	FC_U	DISTURBED_ABLE	FC_P	OCCUPIED_UNABLE	FC_U	OCCUPIED_UNABLE	FC_P	DISTURBED_UNABLE	FC_U	DISTURBED_UNABLE	FC_P
State	FC_mode																						
OCCUPIED_ABLE	FC_C																						
OCCUPIED_ABLE	FC_U																						
OCCUPIED_ABLE	FC_P																						
DISTURBED_ABLE	FC_C																						
DISTURBED_ABLE	FC_U																						
DISTURBED_ABLE	FC_P																						
OCCUPIED_UNABLE	FC_U																						
OCCUPIED_UNABLE	FC_P																						
DISTURBED_UNABLE	FC_U																						
DISTURBED_UNABLE	FC_P																						

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Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-AC:SSS:REQ: 2131

Axle Counter - Reject reset command

Given the current element state is <State>
And the Axle Counter element is configured to support the <FC_mode> mode of the reset command
When receiving the command e_reset_rp from CC
Then SEA shall report the element state <CCState> with command reply flag true to CC
And SEA shall report the rejected command diagnosis to DGP

State	FC_mode	CCState
FREE	FC_C	e_track_free_es
FREE	FC_U	e_track_free_es
FREE	FC_P	e_track_free_es
UNDEFINED	FC_C	e_undefined_es
UNDEFINED	FC_U	e_undefined_es
UNDEFINED	FC_P	e_undefined_es
OCCUPIED_TIMED	FC_C	e_track_occupied_es
OCCUPIED_TIMED	FC_U	e_track_occupied_es
OCCUPIED_TIMED	FC_P	e_track_occupied_es
DISTURBED_TIMED	FC_C	e_track_not_regular_es
DISTURBED_TIMED	FC_U	e_track_not_regular_es
		e_track_not_regular_

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State	FC_mode	CCState
DISTURBED_TIMED	FC_P	es
OCCUPIED_UNABLE	FC_C	e_track_occupied_es
DISTURBED_UNABLE	FC_C	e_track_not_regular_es

Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:	Receiving the command e_reset_rp in state UNDEFINED means, that it has been received after establishing the TDS connection and TDS Occupancy Status is one of WAITING_FOR_SWEEPING_TRAIN, SWEEPING_TRAIN_DETECTED.		
Satisfies:			

EC-MUX-AC:SSS:REQ: 2132

Axle Counter - command reply with message Command Rejected from TDS

Given an Axle Counter element with an open FC command with <FC_mode> mode
And the current element state is <State>

When receiving the message Command Rejected with Reason <RejectReason> from TDS

Then SEA shall report the element state <CCState> with command reply flag true to CC
And SEA shall report the plain text diagnosis message "Command <FC_mode> rejected by TDS for AC x" to DGP

State	FC_mode	RejectReason	CCState
OCCUPIED_ ABLE	FC_ C	OPERATIONAL	e_track_ occupied_ es
OCCUPIED_ ABLE	FC_ U	OPERATIONAL	e_track_ occupied_ es
OCCUPIED_ ABLE	FC_ P	OPERATIONAL	e_track_ occupied_ es
DISTURBED_ ABLE	FC_ C	OPERATIONAL	e_track_ not_ regular_ es

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State	FC_mode	RejectReason	CCState
DISTURBED_ ABLE	FC_ U	OPERATIONAL	e_ track_ not_ regular_ es
DISTURBED_ ABLE	FC_ P	OPERATIONAL	e_ track_ not_ regular_ es
OCCUPIED_ UNABLE	FC_ U	OPERATIONAL	e_ track_ occupied_ es
OCCUPIED_ UNABLE	FC_ P	OPERATIONAL	e_ track_ occupied_ es
DISTURBED_ UNABLE	FC_ U	OPERATIONAL	e_ track_ not_ regular_ es
DISTURBED_ UNABLE	FC_ P	OPERATIONAL	e_ track_ not_ regular_ es
OCCUPIED_ ABLE	FC_ C	TECHNICAL	e_ track_ occupied_ es
OCCUPIED_ ABLE	FC_ U	TECHNICAL	e_ track_ occupied_ es
OCCUPIED_ ABLE	FC_ P	TECHNICAL	e_ track_ occupied_ es
DISTURBED_ ABLE	FC_ C	TECHNICAL	e_ track_ not_ regular_ es
DISTURBED_ ABLE	FC_ U	TECHNICAL	e_ track_ not_ regular_ es
DISTURBED_ ABLE	FC_ P	TECHNICAL	e_ track_ not_ regular_ es
OCCUPIED_ UNABLE	FC_ U	TECHNICAL	e_ track_ occupied_ es
OCCUPIED_ UNABLE	FC_ P	TECHNICAL	e_ track_ occupied_ es
DISTURBED_ UNABLE	FC_ U	TECHNICAL	e_ track_ not_ regular_ es
DISTURBED_ UNABLE	FC_ P	TECHNICAL	e_ track_ not_ regular_ es

Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

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EC-MUX-AC:SSS:REQ: 2133

Axle Counter - command reply with message TVPS FC_P Failed

Given an Axle Counter element with an open FC command with FC_P mode
And the current element state is <State>

When receiving the message TVPS FC_P Failed with Reason <FailReason> from TDS

Then SEA shall report the element state <CCState> with command reply flag true to CC
And SEA shall report the plain text diagnosis message "Command FC_P rejected by TDS for AC x" to DGP

State	FailReason	CCState
OCCUPIED_ UNABLE	INCORRECT_ COUNT	e_ track_ occupied_ es
DISTURBED_ UNABLE	INCORRECT_ COUNT	e_ track_ not_ regular_ es
OCCUPIED_ ABLE	INCORRECT_ COUNT	e_ track_ occupied_ es
DISTURBED_ ABLE	INCORRECT_ COUNT	e_ track_ not_ regular_ es
OCCUPIED_ UNABLE	TIMEOUT_ EXPIRED	e_ track_ occupied_ es
DISTURBED_ UNABLE	TIMEOUT_ EXPIRED	e_ track_ not_ regular_ es
OCCUPIED_ ABLE	TIMEOUT_ EXPIRED	e_ track_ occupied_ es
DISTURBED_ ABLE	TIMEOUT_ EXPIRED	e_ track_ not_ regular_ es
OCCUPIED_ UNABLE	BOUNDING_ NOT_ PERMITTED	e_ track_ occupied_ es
DISTURBED_ UNABLE	BOUNDING_ NOT_ PERMITTED	e_ track_ not_ regular_ es
OCCUPIED_ ABLE	BOUNDING_ NOT_ PERMITTED	e_ track_ occupied_ es
DISTURBED_ ABLE	BOUNDING_ NOT_ PERMITTED	e_ track_ not_ regular_ es
OCCUPIED_ UNABLE	INTENTIONALLY_ DELETED	e_ track_ occupied_ es
DISTURBED_ UNABLE	INTENTIONALLY_ DELETED	e_ track_ not_ regular_ es
OCCUPIED_ ABLE	INTENTIONALLY_ DELETED	e_ track_ occupied_ es

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State	FailReason	CCState
DISTURBED_ ABLE	INTENTIONALLY_ DELETED	e_track_not_regular_es
OCCUPIED_ UNABLE	OUTGOING_ AXLE_ TO_ EXP	e_track_occupied_es
DISTURBED_ UNABLE	OUTGOING_ AXLE_ TO_ EXP	e_track_not_regular_es
OCCUPIED_ ABLE	OUTGOING_ AXLE_ TO_ EXP	e_track_occupied_es
DISTURBED_ ABLE	OUTGOING_ AXLE_ TO_ EXP	e_track_not_regular_es
OCCUPIED_ UNABLE	PROCESS_ CANCELLED	e_track_occupied_es
DISTURBED_ UNABLE	PROCESS_ CANCELLED	e_track_not_regular_es
OCCUPIED_ ABLE	PROCESS_ CANCELLED	e_track_occupied_es
DISTURBED_ ABLE	PROCESS_ CANCELLED	e_track_not_regular_es

Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-AC:SSS:REQ: 2134	Axle Counter - Spontaneous message - Disconnect on Command Rejected in UNDEFINED
Given the current element state is UNDEFINED When receiving the message Command Rejected with Reason <RejectReason> from TDS Then SEA shall report the plain text diagnosis message "TDS-IF-ERROR: Invalid sequence - spontaneous message Command Rejected for AC x" to DGP And SEA shall execute EC-MUX safety reaction OC Disconnection And SEA shall report the Axle Counter no_connection diagnosis device fault message to DGP	

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Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:	Receiving the message Command Rejected in state UNDEFINED means, that it has been received after establishing the TDS connection and TDS Occupancy Status is one of WAITING_FOR_SWEEPING_TRAIN, SWEEPING_TRAIN_DETECTED.		
Satisfies:			

EC-MUX-AC:SSS:REQ: 2135	Axle Counter - Spontaneous message - do nothing on Command Rejected																								
Given an Axle Counter element with no open reset command And the current element state is <State> When receiving the message Command Rejected with Reason <RejectReason> from TDS Then SEA shall report the plain text diagnosis message "TDS-IF-ERROR: Invalid sequence - spontaneous message Command Rejected for AC x" to DGP <table border="1"> <thead> <tr> <th>State</th><th>RejectReason</th></tr> </thead> <tbody> <tr><td>FREE</td><td>OPERATIONAL</td></tr> <tr><td>OCCUPIED_ ABLE</td><td>OPERATIONAL</td></tr> <tr><td>OCCUPIED_ UNABLE</td><td>OPERATIONAL</td></tr> <tr><td>OCCUPIED_ TIMED</td><td>OPERATIONAL</td></tr> <tr><td>DISTURBED_ ABLE</td><td>OPERATIONAL</td></tr> <tr><td>DISTURBED_ UNABLE</td><td>OPERATIONAL</td></tr> <tr><td>DISTURBED_ TIMED</td><td>OPERATIONAL</td></tr> <tr><td>FREE</td><td>TECHNICAL</td></tr> <tr><td>OCCUPIED_ ABLE</td><td>TECHNICAL</td></tr> <tr><td>OCCUPIED_ UNABLE</td><td>TECHNICAL</td></tr> <tr><td>OCCUPIED_ TIMED</td><td>TECHNICAL</td></tr> </tbody> </table>		State	RejectReason	FREE	OPERATIONAL	OCCUPIED_ ABLE	OPERATIONAL	OCCUPIED_ UNABLE	OPERATIONAL	OCCUPIED_ TIMED	OPERATIONAL	DISTURBED_ ABLE	OPERATIONAL	DISTURBED_ UNABLE	OPERATIONAL	DISTURBED_ TIMED	OPERATIONAL	FREE	TECHNICAL	OCCUPIED_ ABLE	TECHNICAL	OCCUPIED_ UNABLE	TECHNICAL	OCCUPIED_ TIMED	TECHNICAL
State	RejectReason																								
FREE	OPERATIONAL																								
OCCUPIED_ ABLE	OPERATIONAL																								
OCCUPIED_ UNABLE	OPERATIONAL																								
OCCUPIED_ TIMED	OPERATIONAL																								
DISTURBED_ ABLE	OPERATIONAL																								
DISTURBED_ UNABLE	OPERATIONAL																								
DISTURBED_ TIMED	OPERATIONAL																								
FREE	TECHNICAL																								
OCCUPIED_ ABLE	TECHNICAL																								
OCCUPIED_ UNABLE	TECHNICAL																								
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	305	16282~9950	ED01 PD01

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State	RejectReason
DISTURBED_ ABLE	TECHNICAL
DISTURBED_ UNABLE	TECHNICAL
DISTURBED_ TIMED	TECHNICAL

Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-AC:SSS:REQ: 2150	Axle Counter - reject repeated command
--------------------------------	---

Given an Axle Counter element with an open FC command with <FC_mode> mode
And the current element state is <State>

When receiving the command e_reset_rp from CC

Then SEA shall report the element state <CCState> with command reply flag true to CC
And SEA shall report the rejected command diagnosis to DGP

State	FC_mode	CCState
OCCUPIED_ ABLE	FC_ C	e_track_occupied_es
OCCUPIED_ ABLE	FC_ U	e_track_occupied_es
OCCUPIED_ ABLE	FC_ P	e_track_occupied_es
DISTURBED_ ABLE	FC_ C	e_track_not_regular_es
DISTURBED_ ABLE	FC_ U	e_track_not_regular_es
		e_track_not_regular_

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State	FC_mode	CCState
DISTURBED_ ABLE	FC_ P	es
OCCUPIED_ UNABLE	FC_ U	e_track_ occupied_ es
OCCUPIED_ UNABLE	FC_ P	e_track_ occupied_ es
DISTURBED_ UNABLE	FC_ U	e_track_ not_ regular_ es
DISTURBED_ UNABLE	FC_ P	e_track_ not_ regular_ es

Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-AC:SSS:REQ: 2151	Axle Counter - Command Reply - Disconnect on TVPS FC_P_A Failed message				
Given an Axle Counter element with an open FC command with <FC_mode> mode When receiving the message TVPS FC_P_A Failed from TDS Then SEA shall report the plain text diagnosis message "TDS-IF-ERROR: Unexpected TVPS FC_P_A Failed message for AC x" to DGP And SEA shall execute EC-MUX safety reaction OC Disconnection And SEA shall report the Axle Counter no_connection diagnosis device fault message to DGP <table><tr><th>FC_mode</th></tr><tr><td>FC_C</td></tr><tr><td>FC_U</td></tr><tr><td>FC_P</td></tr></table>		FC_mode	FC_C	FC_U	FC_P
FC_mode					
FC_C					
FC_U					
FC_P					

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Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-AC:SSS:REQ: 2159	Axle Counter - Spontaneous message - Disconnect on TVPS FC_P_A Failed message		
Given an Axle Counter element with no open reset command When receiving the message TVPS FC_P_A Failed from TDS Then SEA shall report the plain text diagnosis message "TDS-IF-ERROR: Unexpected TVPS FC_P_A Failed message for AC x" to DGP And SEA shall execute EC-MUX safety reaction OC Disconnection And SEA shall report the Axle Counter no_connection diagnosis device fault message to DGP			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-AC:SSS:REQ: 2152	Axle Counter - Command Reply - Disconnect on TVPS FC_P Failed message
Given an Axle Counter element with an open FC command with <FC_mode> mode When receiving the message TVPS FC_P Failed with Reason <FailReason> from TDS	

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Then SEA shall report the plain text diagnosis message "TDS-IF-ERROR: Unexpected TVPS FC_P Failed message for AC x" to DGP
And SEA shall execute EC-MUX safety reaction OC Disconnection
And SEA shall report the Axle Counter no_connection diagnosis device fault message to DGP

FC_mode	FailReason
FC_C	INCORRECT_COUNT
FC_C	TIMEOUT_EXPIRED
FC_C	BOUNDING_NOT_PERMITTED
FC_C	INTENTIONALLY_DELETED
FC_C	OUTGOING_AXLE_TO_EXP
FC_C	PROCESS_CANCELLED
FC_U	INCORRECT_COUNT
FC_U	TIMEOUT_EXPIRED
FC_U	BOUNDING_NOT_PERMITTED
FC_U	INTENTIONALLY_DELETED
FC_U	OUTGOING_AXLE_TO_EXP
FC_U	PROCESS_CANCELLED

Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-AC:SSS:REQ: 2153	Axle Counter - Spontaneous Message - Disconnect on TVPS FC_P Failed message
Given the current element state is <State>	

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When receiving the message TVPS FC_P Failed with Reason <FailReason> from TDS

Then SEA shall report the plain text diagnosis message "TDS-IF-ERROR: Unexpected TVPS FC_P Failed message for AC x" to DGP

And SEA shall execute EC-MUX safety reaction OC Disconnection

And SEA shall report the Axle Counter no_connection diagnosis device fault message to DGP

State	FailReason
OCCUPIED_ ABLE	INCORRECT_ COUNT
OCCUPIED_ ABLE	TIMEOUT_ EXPIRED
OCCUPIED_ ABLE	BOUNDING_ NOT_ PERMITTED
OCCUPIED_ ABLE	INTENTIONALLY_ DELETED
OCCUPIED_ ABLE	OUTGOING_ AXLE_ TO_ EXP
OCCUPIED_ ABLE	PROCESS_ CANCELLED
OCCUPIED_ UNABLE	INCORRECT_ COUNT
OCCUPIED_ UNABLE	TIMEOUT_ EXPIRED
OCCUPIED_ UNABLE	BOUNDING_ NOT_ PERMITTED
OCCUPIED_ UNABLE	INTENTIONALLY_ DELETED
OCCUPIED_ UNABLE	OUTGOING_ AXLE_ TO_ EXP
OCCUPIED_ UNABLE	PROCESS_ CANCELLED
OCCUPIED_ TIMED	INCORRECT_ COUNT
OCCUPIED_ TIMED	TIMEOUT_ EXPIRED
OCCUPIED_ TIMED	BOUNDING_ NOT_ PERMITTED
OCCUPIED_ TIMED	INTENTIONALLY_ DELETED
OCCUPIED_ TIMED	OUTGOING_ AXLE_ TO_ EXP
OCCUPIED_ TIMED	PROCESS_ CANCELLED
DISTURBED_ ABLE	INCORRECT_ COUNT
DISTURBED_ ABLE	TIMEOUT_ EXPIRED
DISTURBED_ ABLE	BOUNDING_ NOT_ PERMITTED
DISTURBED_ ABLE	INTENTIONALLY_ DELETED
DISTURBED_ ABLE	OUTGOING_ AXLE_ TO_ EXP
DISTURBED_ ABLE	PROCESS_ CANCELLED

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State	FailReason
DISTURBED_UNABLE	INCORRECT_COUNT
DISTURBED_UNABLE	TIMEOUT_EXPIRED
DISTURBED_UNABLE	BOUNDING_NOT_PERMITTED
DISTURBED_UNABLE	INTENTIONALLY_DELETED
DISTURBED_UNABLE	OUTGOING_AXLE_TO_EXP
DISTURBED_UNABLE	PROCESS_CANCELLED
DISTURBED_TIMED	INCORRECT_COUNT
DISTURBED_TIMED	TIMEOUT_EXPIRED
DISTURBED_TIMED	BOUNDING_NOT_PERMITTED
DISTURBED_TIMED	INTENTIONALLY_DELETED
DISTURBED_TIMED	OUTGOING_AXLE_TO_EXP
DISTURBED_TIMED	PROCESS_CANCELLED
FREE	INCORRECT_COUNT
FREE	TIMEOUT_EXPIRED
FREE	BOUNDING_NOT_PERMITTED
FREE	INTENTIONALLY_DELETED
FREE	OUTGOING_AXLE_TO_EXP
FREE	PROCESS_CANCELLED
UNDEFINED	INCORRECT_COUNT
UNDEFINED	TIMEOUT_EXPIRED
UNDEFINED	BOUNDING_NOT_PERMITTED
UNDEFINED	INTENTIONALLY_DELETED
UNDEFINED	OUTGOING_AXLE_TO_EXP
UNDEFINED	PROCESS_CANCELLED

Type: FUR - Functional Requirement

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Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:	For UNDEFINED state the message TVPS FC_P Failed has been received after establishing the TDS connection and TDS Occupancy Status is one of WAITING_FOR_SWEEPING_TRAIN, SWEEPING_TRAIN_DETECTED.		
Satisfies:			

3.1.1.2 Operational - Axle Counter - Message

In order to control the element state of the Axle Counter

As a SEA

I want to be able to receive state messages from TDS and report them to CC, including the information about command reply / spontaneous notification.

For requirement language see Chapter 2.6 "Requirement Language".

Given SEA with a configured Axle Counter element

And DelayToFree is the configured time for the spontaneous transition to free state without reset command

And the mapping between SEA diagnosis aspect and DGP diagnosis value is defined

SEA diagnosis aspect	DGP diagnosis value
AC_FREE	0
AC_OCCUPIED_ABLE	1
AC_OCCUPIED_UNABLE	2
AC_OCCUPIED_TIMED	3
AC_DISTURBED_ABLE	4
AC_DISTURBED_UNABLE	5
AC_DISTURBED_TIMED	6
AC_UNDEFINED	7

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EC-MUX-AC:SSS:REQ: 2136	Axle Counter - initial state		
When initializing the element			
Then SEA shall set the current element state to UNDEFINED			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test
Comments:			
Satisfies:			

EC-MUX-AC:SSS:REQ: 2137	Axle Counter - Spontaneous message - State transition with reporting without fault information from FREE											
Given an Axle Counter element with no open reset command And the current element state is FREE And status message contains Occupancy Status OCCUPIED And status message contains Force Ability <Ability> When receiving the status message from TDS Then SEA shall set the current element state to <NewState> And SEA shall report the element state e_track_occupied_es with command reply flag false to CC And SEA shall report the diagnosis aspect message <DGPMMessage> to DGP <table><tr><th>Ability</th><th>NewState</th><th>DGPMMessage</th></tr><tr><td>ABLE</td><td>OCCUPIED_ ABLE</td><td>AC_ OCCUPIED_ ABLE</td></tr><tr><td>UNABLE</td><td>OCCUPIED_ UNABLE</td><td>AC_ OCCUPIED_ UNABLE</td></tr></table>				Ability	NewState	DGPMMessage	ABLE	OCCUPIED_ ABLE	AC_ OCCUPIED_ ABLE	UNABLE	OCCUPIED_ UNABLE	AC_ OCCUPIED_ UNABLE
Ability	NewState	DGPMMessage										
ABLE	OCCUPIED_ ABLE	AC_ OCCUPIED_ ABLE										
UNABLE	OCCUPIED_ UNABLE	AC_ OCCUPIED_ UNABLE										
Type:	FUR - Functional Requirement											
Status:	draft	Stability:	stable									
Target Release:	SEA SW GEN 3.0	Priority:	mandatory									

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Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-AC:SSS:REQ: 2188	Axle Counter - Spontaneous message - Ignore repeated VACANT message in FREE		
Given an Axle Counter element with no open reset command And the current element state is FREE And status message contains Occupancy Status VACANT And status message contains Force Ability UNABLE When receiving the status message from TDS Then SEA shall do nothing			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-AC:SSS:REQ: 2187	Axle Counter - Spontaneous message - State transition with reporting without fault information from UNDEFINED
Given an Axle Counter element with no open reset command And the current element state is UNDEFINED And TDS Occupancy Status is <OccupStatus> And TDS Force Ability is UNABLE And status message contains Occupancy Status DISTURBED And status message contains Force Ability <Ability>	

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When receiving the status message from TDS																							
Then SEA shall set the current element state to <NewState>																							
And SEA shall report the element state e_track_not_regular_es with command reply flag false to CC																							
And SEA shall report the diagnosis aspect message <DGPMMessage> to DGP																							
<table><tr><th>OccupStatus</th><th>Ability</th><th>NewState</th><th>DGPMMessage</th></tr><tr><td>SWEEPING_ TRAIN_ DETECTED</td><td>ABLE</td><td>DISTURBED_ ABLE</td><td>AC_ DISTURBED_ ABLE</td></tr><tr><td>WAITING_ FOR_ SWEEPING_ TRAIN</td><td>ABLE</td><td>DISTURBED_ ABLE</td><td>AC_ DISTURBED_ ABLE</td></tr><tr><td>SWEEPING_ TRAIN_ DETECTED</td><td>UNABLE</td><td>DISTURBED_ UNABLE</td><td>AC_ DISTURBED_ UNABLE</td></tr><tr><td>WAITING_ FOR_ SWEEPING_ TRAIN</td><td>UNABLE</td><td>DISTURBED_ UNABLE</td><td>AC_ DISTURBED_ UNABLE</td></tr></table>				OccupStatus	Ability	NewState	DGPMMessage	SWEEPING_ TRAIN_ DETECTED	ABLE	DISTURBED_ ABLE	AC_ DISTURBED_ ABLE	WAITING_ FOR_ SWEEPING_ TRAIN	ABLE	DISTURBED_ ABLE	AC_ DISTURBED_ ABLE	SWEEPING_ TRAIN_ DETECTED	UNABLE	DISTURBED_ UNABLE	AC_ DISTURBED_ UNABLE	WAITING_ FOR_ SWEEPING_ TRAIN	UNABLE	DISTURBED_ UNABLE	AC_ DISTURBED_ UNABLE
OccupStatus	Ability	NewState	DGPMMessage																				
SWEEPING_ TRAIN_ DETECTED	ABLE	DISTURBED_ ABLE	AC_ DISTURBED_ ABLE																				
WAITING_ FOR_ SWEEPING_ TRAIN	ABLE	DISTURBED_ ABLE	AC_ DISTURBED_ ABLE																				
SWEEPING_ TRAIN_ DETECTED	UNABLE	DISTURBED_ UNABLE	AC_ DISTURBED_ UNABLE																				
WAITING_ FOR_ SWEEPING_ TRAIN	UNABLE	DISTURBED_ UNABLE	AC_ DISTURBED_ UNABLE																				
Type:	FUR - Functional Requirement																						
Status:	draft	Stability:	stable																				
Target Release:	SEA SW GEN 3.0	Priority:	mandatory																				
Originator:	Technics (R&D)	Allocation:	SW EC-MUX																				
Verification Method:	Test	Verification Step:	module test, subit																				
Comments:	Receiving the the status message in state UNDEFINED means, that it has been received after establishing the TDS connection and TDS Occupancy Status is one of WAITING_FOR_SWEEPING_TRAIN, SWEEPING_TRAIN_DETECTED.																						
Satisfies:																							

EC-MUX-AC:SSS:REQ: 2138	Axle Counter - Spontaneous message - State transition with reporting with fault information
Given an Axle Counter element with no open reset command	
And the current element state is <State>	
And status message contains Occupancy Status <OccupStatus>	
And status message contains Force Ability <Ability>	
When receiving the status message from TDS	

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Then SEA shall set the current element state to <NewState>

And SEA shall report the element state <CCState> with command reply flag false to CC

And SEA shall send the fault notification <FaultMsg> with fault type e_if_error to CC

And SEA shall report the diagnosis aspect message <DGPMMessage> to DGP

State	OccupStatus	Ability	NewState	CCState	FaultMsg	DGPMMessage
DISTURBED_ABLE	OCCUPIED	ABLE	OCCUPIED_ABLE	e_track_occupied_es	sum_fault_rep	AC_OCCUPIED_ABLE
DISTURBED_ABLE	OCCUPIED	UNABLE	OCCUPIED_UNABLE	e_track_occupied_es	sum_fault_rep	AC_OCCUPIED_UNABLE
DISTURBED_UNABLE	OCCUPIED	UNABLE	OCCUPIED_UNABLE	e_track_occupied_es	sum_fault_rep	AC_OCCUPIED_UNABLE
DISTURBED_UNABLE	OCCUPIED	ABLE	OCCUPIED_ABLE	e_track_occupied_es	sum_fault_rep	AC_OCCUPIED_ABLE
FREE	DISTURBED	ABLE	DISTURBED_ABLE	e_track_not_regular_es	sum_fault	AC_DISTURBED_ABLE
FREE	DISTURBED	UNABLE	DISTURBED_UNABLE	e_track_not_regular_es	sum_fault	AC_DISTURBED_UNABLE
OCCUPIED_ABLE	DISTURBED	ABLE	DISTURBED_ABLE	e_track_not_regular_es	sum_fault	AC_DISTURBED_ABLE
OCCUPIED_ABLE	DISTURBED	UNABLE	DISTURBED_UNABLE	e_track_not_regular_es	sum_fault	AC_DISTURBED_UNABLE
OCCUPIED_UNABLE	DISTURBED	ABLE	DISTURBED_ABLE	e_track_not_regular_es	sum_fault	AC_DISTURBED_ABLE
OCCUPIED_UNABLE	DISTURBED	UNABLE	DISTURBED_UNABLE	e_track_not_regular_es	sum_fault	AC_DISTURBED_UNABLE

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Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-AC:SSS:REQ: 2186	Axle Counter - Spontaneous message - State transition with reporting with fault information from UNDEFINED				
Given an Axle Counter element with no open reset command And the current element state is UNDEFINED And TDS Occupancy Status is <InitOccupStatus> And TDS Force Ability is UNABLE And status message contains Occupancy Status <OccupStatus> And status message contains Force Ability <Ability> When receiving the status message from TDS Then SEA shall set the current element state to <NewState> And SEA shall report the element state <CCState> with command reply flag false to CC And SEA shall send the fault notification sum_fault_rep with fault type e_if_error to CC And SEA shall report the diagnosis aspect message <DGPMMessage> to DGP					
InitOccupStatus	OccupStatus	Ability	NewState	CCState	DGPMMessage
SWEEPING_TRAIN_DETECTED	VACANT	UNABLE	FREE	e_track_free_es	AC_FREE
WAITING_FOR_SWEEPING_TRAIN	VACANT	UNABLE	FREE	e_track_free_es	AC_FREE
SWEEPING_TRAIN_DETECTED	OCCUPIED	ABLE	OCCUPIED_ABLE	e_track_occupied_es	AC_OCCUPIED_ABLE
WAITING_FOR_SWEEPING_TRAIN	OCCUPIED	ABLE	OCCUPIED_ABLE	e_track_occupied_es	AC_OCCUPIED_ABLE
				e_track_occupied_	AC_OCCUPIED_

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InitOccupStatus	OccupStatus	Ability	NewState	CCState	DGPMessage
SWEEPING_TRAIN_DETECTED	OCCUPIED	UNABLE	OCCUPIED_UNABLE	es	UNABLE
WAITING_FOR_SWEEPING_TRAIN	OCCUPIED	UNABLE	OCCUPIED_UNABLE	e_track_occupied_es	AC_OCCUPIED_UNABLE

Type:	FUR - Functional Requirement			
Status:	draft	Stability:	stable	
Target Release:	SEA SW GEN 3.0	Priority:	mandatory	
Originator:	Technics (R&D)	Allocation:	SW EC-MUX	
Verification Method:	Test	Verification Step:	module test, subit	
Comments:	Receiving the the status message in state UNDEFINED means, that it has been received after establishing the TDS connection and TDS Occupancy Status is one of WAITING_FOR_SWEEPING_TRAIN, SWEEPING_TRAIN_DETECTED.			
Satisfies:				

EC-MUX-AC:SSS:REQ: 2161	Axle Counter - Spontaneous message - Ignore repeated sweeping train messages in UNDEFINED
--------------------------------	--

Given an Axle Counter element with no open reset command
And the current element state is UNDEFINED
And TDS Occupancy Status is <InitOccupStatus>
And TDS Force Ability is UNABLE
And status message contains Occupancy Status <OccupStatus>
And status message contains Force Ability UNABLE

When receiving the status message from TDS

Then SEA shall do nothing

InitOccupStatus	OccupStatus
SWEEPING_TRAIN_DETECTED	WAITING_FOR_SWEEPING_TRAIN
WAITING_FOR_SWEEPING_TRAIN	SWEEPING_TRAIN_DETECTED

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Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:	Receiving the the status message in state UNDEFINED means, that it has been received after establishing the TDS connection and TDS Occupancy Status is one of WAITING_FOR_SWEEPING_TRAIN, SWEEPING_TRAIN_DETECTED.		
Satisfies:			

EC-MUX-AC:SSS:REQ: 2139	Axle Counter - Spontaneous message - State transition without reporting and without fault information																												
Given an Axle Counter element with no open reset command And the current element state is <State> And status message contains Occupancy Status <OccupStatus> And status message contains Force Ability <Ability> When receiving the status message from TDS Then SEA shall set the current element state to <NewState> And SEA shall report the diagnosis aspect message <DGPMMessage> to DGP <table><tr><th>State</th><th>OccupStatus</th><th>Ability</th><th>NewState</th><th>DGPMessage</th></tr><tr><td>DISTURBED_ ABLE</td><td>DISTURBED</td><td>UNABLE</td><td>DISTURBED_ UNABLE</td><td>AC_ DISTURBED_ UNABLE</td></tr><tr><td>DISTURBED_ UNABLE</td><td>DISTURBED</td><td>ABLE</td><td>DISTURBED_ ABLE</td><td>AC_ DISTURBED_ ABLE</td></tr><tr><td>OCCUPIED_ ABLE</td><td>OCCUPIED</td><td>UNABLE</td><td>OCCUPIED_ UNABLE</td><td>AC_ OCCUPIED_ UNABLE</td></tr><tr><td>OCCUPIED_ UNABLE</td><td>OCCUPIED</td><td>ABLE</td><td>OCCUPIED_ ABLE</td><td>AC_ OCCUPIED_ ABLE</td></tr></table>					State	OccupStatus	Ability	NewState	DGPMessage	DISTURBED_ ABLE	DISTURBED	UNABLE	DISTURBED_ UNABLE	AC_ DISTURBED_ UNABLE	DISTURBED_ UNABLE	DISTURBED	ABLE	DISTURBED_ ABLE	AC_ DISTURBED_ ABLE	OCCUPIED_ ABLE	OCCUPIED	UNABLE	OCCUPIED_ UNABLE	AC_ OCCUPIED_ UNABLE	OCCUPIED_ UNABLE	OCCUPIED	ABLE	OCCUPIED_ ABLE	AC_ OCCUPIED_ ABLE
State	OccupStatus	Ability	NewState	DGPMessage																									
DISTURBED_ ABLE	DISTURBED	UNABLE	DISTURBED_ UNABLE	AC_ DISTURBED_ UNABLE																									
DISTURBED_ UNABLE	DISTURBED	ABLE	DISTURBED_ ABLE	AC_ DISTURBED_ ABLE																									
OCCUPIED_ ABLE	OCCUPIED	UNABLE	OCCUPIED_ UNABLE	AC_ OCCUPIED_ UNABLE																									
OCCUPIED_ UNABLE	OCCUPIED	ABLE	OCCUPIED_ ABLE	AC_ OCCUPIED_ ABLE																									
Type:	FUR - Functional Requirement																												
Status:	draft		Stability:	stable																									
Target Release:	SEA SW GEN 3.0		Priority:	mandatory																									

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	305	16282-9950	ED01 PD01

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Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-AC:SSS:REQ: 2160	Axle Counter - Spontaneous message - Ignore repeated messages																	
Given an Axle Counter element with no open reset command And the current element state is <State> And status message contains Occupancy Status <OccupStatus> And status message contains Force Ability <Ability> When receiving the status message from TDS Then SEA shall do nothing <table><tr><th>State</th><th>OccupStatus</th><th>Ability</th></tr><tr><td>DISTURBED_ ABLE</td><td>DISTURBED</td><td>ABLE</td></tr><tr><td>DISTURBED_ UNABLE</td><td>DISTURBED</td><td>UNABLE</td></tr><tr><td>OCCUPIED_ ABLE</td><td>OCCUPIED</td><td>ABLE</td></tr><tr><td>OCCUPIED_ UNABLE</td><td>OCCUPIED</td><td>UNABLE</td></tr></table>				State	OccupStatus	Ability	DISTURBED_ ABLE	DISTURBED	ABLE	DISTURBED_ UNABLE	DISTURBED	UNABLE	OCCUPIED_ ABLE	OCCUPIED	ABLE	OCCUPIED_ UNABLE	OCCUPIED	UNABLE
State	OccupStatus	Ability																
DISTURBED_ ABLE	DISTURBED	ABLE																
DISTURBED_ UNABLE	DISTURBED	UNABLE																
OCCUPIED_ ABLE	OCCUPIED	ABLE																
OCCUPIED_ UNABLE	OCCUPIED	UNABLE																
Type:	FUR - Functional Requirement																	
Status:	draft	Stability:	stable															
Target Release:	SEA SW GEN 3.0	Priority:	mandatory															
Originator:	Technics (R&D)	Allocation:	SW EC-MUX															
Verification Method:	Test	Verification Step:	module test, subit															
Comments:																		
Satisfies:																		

EC-MUX-AC:SSS:REQ:	Axle Counter - Spontaneous message - State transition to FREE with
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2140	timer					
Given an Axle Counter element with no open reset command And the current element state is <State> And status message contains Occupancy Status VACANT And status message contains Force Ability UNABLE When receiving the status message from TDS Then SEA shall set the current element state to OCCUPIED_TIMED And SEA shall report the diagnosis aspect message AC_OCCUPIED_TIMED to DGP And SEA shall set the current element state to FREE after DelayToFree s And SEA shall report the element state e_track_free_es with command reply flag false to CC And SEA shall report the diagnosis aspect message AC_FREE to DGP <table><tr><th>State</th></tr><tr><td>OCCUPIED_ ABLE</td></tr><tr><td>OCCUPIED_ UNABLE</td></tr></table>				State	OCCUPIED_ ABLE	OCCUPIED_ UNABLE
State						
OCCUPIED_ ABLE						
OCCUPIED_ UNABLE						
Type:	FUR - Functional Requirement					
Status:	draft	Stability:	stable			
Target Release:	SEA SW GEN 3.0	Priority:	mandatory			
Originator:	Technics (R&D)	Allocation:	SW EC-MUX			
Verification Method:	Test	Verification Step:	module test, subit			
Comments:						
Satisfies:						

EC-MUX-AC:SSS:REQ: 2141	Axle Counter - Spontaneous message - State transition to FREE with timer and fault information
Given an Axle Counter element with no open reset command And the current element state is <State> And status message contains Occupancy Status VACANT And status message contains Force Ability UNABLE When receiving the status message from TDS	

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Then SEA shall set the current element state to DISTURBED_TIMED
And SEA shall report the diagnosis aspect message AC_DISTURBED_TIMED to DGP
And SEA shall set the current element state to FREE after DelayToFree s
And SEA shall report the element state e_track_free_es with command reply flag false to CC
And SEA shall send the fault notification sum_fault_rep with fault type e_if_error to CC
And SEA shall report the diagnosis aspect message AC_FREE to DGP

State
DISTURBED_ABLE
DISTURBED_UNABLE

Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-AC:SSS:REQ: 2142	Axle Counter - Spontaneous message - State transition with timer cancelling and fault information					
Given an Axle Counter element with no open reset command And the current element state is <State> And status message contains Occupancy Status <OccupStatus> And status message contains Force Ability <Ability> When receiving the status message from TDS within DelayToFree s Then SEA shall set the current element state to <NewState> And SEA shall report the element state <CCState> with command reply flag false to CC And SEA shall send the fault notification <FaultMsg> with fault type e_if_error to CC And SEA shall report the diagnosis aspect message <DGPMMessage> to DGP						
State	OccupStatus	Ability	NewState	CCState	FaultMsg	DGPMMessage
				e_track_not_		AC_DISTURBED_

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State	OccupStatus	Ability	NewState	CCState	FaultMsg	DGPMessage
OCCUPIED_TIMED	DISTURBED	ABLE	DISTURBED_ABLE	regular_es	sum_fault	ABLE
OCCUPIED_TIMED	DISTURBED	UNABLE	DISTURBED_UNABLE	e_track_not_regular_es	sum_fault	AC_DISTURBED_UNABLE
DISTURBED_TIMED	OCCUPIED	ABLE	OCCUPIED_ABLE	e_track_occupied_es	sum_fault_rep	AC_OCCUPIED_ABLE
DISTURBED_TIMED	OCCUPIED	UNABLE	OCCUPIED_UNABLE	e_track_occupied_es	sum_fault_rep	AC_OCCUPIED_UNABLE

Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-AC:SSS:REQ: 2143	Axle Counter - Spontaneous message - State transition with timer cancelling and without fault information										
Given an Axle Counter element with no open reset command And the current element state is <State> And status message contains Occupancy Status <OccupStatus> And status message contains Force Ability <Ability> When receiving the status message from TDS within DelayToFree s Then SEA shall set the current element state to <NewState> And SEA shall report the diagnosis aspect message <DGPMessage> to DGP											
<table><tr><th>State</th><th>OccupStatus</th><th>Ability</th><th>NewState</th><th>DGPMessage</th></tr><tr><td>OCCUPIED_TIMED</td><td>OCCUPIED</td><td>ABLE</td><td>OCCUPIED_ ABLE</td><td>AC_OCCUPIED_ ABLE</td></tr></table>		State	OccupStatus	Ability	NewState	DGPMessage	OCCUPIED_TIMED	OCCUPIED	ABLE	OCCUPIED_ ABLE	AC_OCCUPIED_ ABLE
State	OccupStatus	Ability	NewState	DGPMessage							
OCCUPIED_TIMED	OCCUPIED	ABLE	OCCUPIED_ ABLE	AC_OCCUPIED_ ABLE							

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State	OccupStatus	Ability	NewState	DGPMessage
OCCUPIED_ TIMED	OCCUPIED	UNABLE	OCCUPIED_ UNABLE	AC_ OCCUPIED_ UNABLE
DISTURBED_ TIMED	DISTURBED	ABLE	DISTURBED_ ABLE	AC_ DISTURBED_ ABLE
DISTURBED_ TIMED	DISTURBED	UNABLE	DISTURBED_ UNABLE	AC_ DISTURBED_ UNABLE

Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-AC:SSS:REQ:2154	Axle Counter - Spontaneous message - Ignore Occupancy Status VACANT in timed state					
Given an Axle Counter element with no open reset command And the current element state is <State> And status message contains Occupancy Status VACANT And status message contains Force Ability UNABLE When receiving the status message from TDS within DelayToFree s Then SEA shall do nothing <table><tr><th>State</th></tr><tr><td>OCCUPIED_ TIMED</td></tr><tr><td>DISTURBED_ TIMED</td></tr></table>				State	OCCUPIED_ TIMED	DISTURBED_ TIMED
State						
OCCUPIED_ TIMED						
DISTURBED_ TIMED						
Type:	FUR - Functional Requirement					
Status:	draft	Stability:	stable			
Target Release:	SEA SW GEN 3.0	Priority:	mandatory			

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Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-AC:SSS:REQ: 2144	Axle Counter - Command reply - State transition after successful FC command with reporting and without fault information														
Given an Axle Counter element with an open FC command with <FC_mode> mode And the current element state is <State> And status message contains Occupancy Status VACANT And status message contains Force Ability UNABLE When receiving the status message from TDS Then SEA shall set the current element state to FREE And SEA shall report the element state e_track_free_es with command reply flag true to CC And SEA shall report the diagnosis aspect message AC_FREE to DGP <table><tr><th>FC_mode</th><th>State</th></tr><tr><td>FC_C</td><td>OCCUPIED_ ABLE</td></tr><tr><td>FC_U</td><td>OCCUPIED_ ABLE</td></tr><tr><td>FC_U</td><td>OCCUPIED_ UNABLE</td></tr><tr><td>FC_P</td><td>OCCUPIED_ ABLE</td></tr><tr><td>FC_P</td><td>OCCUPIED_ UNABLE</td></tr></table>				FC_mode	State	FC_C	OCCUPIED_ ABLE	FC_U	OCCUPIED_ ABLE	FC_U	OCCUPIED_ UNABLE	FC_P	OCCUPIED_ ABLE	FC_P	OCCUPIED_ UNABLE
FC_mode	State														
FC_C	OCCUPIED_ ABLE														
FC_U	OCCUPIED_ ABLE														
FC_U	OCCUPIED_ UNABLE														
FC_P	OCCUPIED_ ABLE														
FC_P	OCCUPIED_ UNABLE														
Type:	FUR - Functional Requirement														
Status:	draft	Stability:	stable												
Target Release:	SEA SW GEN 3.0	Priority:	mandatory												
Originator:	Technics (R&D)	Allocation:	SW EC-MUX												
Verification Method:	Test	Verification Step:	module test, subit												
Comments:															

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Satisfies:	
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EC-MUX-AC:SSS:REQ: 2145	Axle Counter - Command reply - State transition after successful FC command with reporting and with fault information														
Given an Axle Counter element with an open FC command with <FC_mode> mode And the current element state is <State> And status message contains Occupancy Status VACANT And status message contains Force Ability UNABLE When receiving the status message from TDS Then SEA shall set the current element state to FREE And SEA shall report the element state e_track_free_es with command reply flag true to CC And SEA shall send the fault notification sum_fault_rep with fault type e_if_error to CC And SEA shall report the diagnosis aspect message AC_FREE to DGP <table><tr><th>FC_mode</th><th>State</th></tr><tr><td>FC_C</td><td>DISTURBED_ ABLE</td></tr><tr><td>FC_U</td><td>DISTURBED_ UNABLE</td></tr><tr><td>FC_U</td><td>DISTURBED_ ABLE</td></tr><tr><td>FC_P</td><td>DISTURBED_ UNABLE</td></tr><tr><td>FC_P</td><td>DISTURBED_ ABLE</td></tr></table>				FC_mode	State	FC_C	DISTURBED_ ABLE	FC_U	DISTURBED_ UNABLE	FC_U	DISTURBED_ ABLE	FC_P	DISTURBED_ UNABLE	FC_P	DISTURBED_ ABLE
FC_mode	State														
FC_C	DISTURBED_ ABLE														
FC_U	DISTURBED_ UNABLE														
FC_U	DISTURBED_ ABLE														
FC_P	DISTURBED_ UNABLE														
FC_P	DISTURBED_ ABLE														
Type:	FUR - Functional Requirement														
Status:	draft	Stability:	stable												
Target Release:	SEA SW GEN 3.0	Priority:	mandatory												
Originator:	Technics (R&D)	Allocation:	SW EC-MUX												
Verification Method:	Test	Verification Step:	module test, subit												
Comments:															
Satisfies:															

EC-MUX-AC:SSS:REQ:	Axle Counter - Command reply - State transition with reporting without
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2146

fault information

Given an Axle Counter element with an open FC command with <FC_mode> mode
And the current element state is <State>
And status message contains Occupancy Status <OccupStatus>
And status message contains Force Ability <Ability>

When receiving the status message from TDS

Then SEA shall set the current element state to <NewState>
And SEA shall report the element state <CCState> with command reply flag true to CC
And SEA shall report the diagnosis aspect message <DGPMMessage> to DGP

FC_mode	State	OccupStatus	Ability	NewState	CCState	DGPMMessage
FC_U	OCCUPIED_ABLE	OCCUPIED	UNABLE	OCCUPIED_UNABLE	e_track_occupied_es	AC_OCCUPIED_UNABLE
FC_P	OCCUPIED_ABLE	OCCUPIED	UNABLE	OCCUPIED_UNABLE	e_track_occupied_es	AC_OCCUPIED_UNABLE
FC_C	OCCUPIED_ABLE	OCCUPIED	UNABLE	OCCUPIED_UNABLE	e_track_occupied_es	AC_OCCUPIED_UNABLE
FC_U	OCCUPIED_UNABLE	OCCUPIED	ABLE	OCCUPIED_ABLE	e_track_occupied_es	AC_OCCUPIED_ABLE
FC_P	OCCUPIED_UNABLE	OCCUPIED	ABLE	OCCUPIED_ABLE	e_track_occupied_es	AC_OCCUPIED_ABLE
FC_U	DISTURBED_ABLE	DISTURBED	UNABLE	DISTURBED_UNABLE	e_track_not_regular_es	AC_DISTURBED_UNABLE
FC_P	DISTURBED_ABLE	DISTURBED	UNABLE	DISTURBED_UNABLE	e_track_not_regular_es	AC_DISTURBED_UNABLE
FC_C	DISTURBED_ABLE	DISTURBED	UNABLE	DISTURBED_UNABLE	e_track_not_regular_es	AC_DISTURBED_UNABLE
FC_U	DISTURBED_UNABLE	DISTURBED	ABLE	DISTURBED_ABLE	e_track_not_regular_es	AC_DISTURBED_ABLE
FC_	DISTURBED_			DISTURBED_	e_track_	AC_

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FC_mode	State	OccupStatus	Ability	NewState	CCState	DGPMessage
P	UNABLE	DISTURBED	ABLE	ABLE	not_regular_es	DISTURBED_ABLE

Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-AC:SSS:REQ: 2147 Axle Counter - Command reply - State transition with reporting with fault occurrence information

Given an Axle Counter element with an open FC command with <FC_mode> mode

And the current element state is <State>

And status message contains Occupancy Status DISTURBED

And status message contains Force Ability <Ability>

When receiving the status message from TDS

Then SEA shall set the current element state to <NewState>

And SEA shall report the element state e_track_not_regular_es with command reply flag true to CC

And SEA shall send the fault notification sum_fault with fault type e_if_error to CC

And SEA shall report the diagnosis aspect message <DGPMessage> to DGP

FC_mode	State	Ability	NewState	DGPMessage
FC_U	OCCUPIED_ABLE	ABLE	DISTURBED_ABLE	AC_DISTURBED_ABLE
FC_P	OCCUPIED_ABLE	ABLE	DISTURBED_ABLE	AC_DISTURBED_ABLE
FC_C	OCCUPIED_ABLE	ABLE	DISTURBED_ABLE	AC_DISTURBED_ABLE
FC_U	OCCUPIED_ABLE	UNABLE	DISTURBED_UNABLE	AC_DISTURBED_UNABLE
FC_P	OCCUPIED_ABLE	UNABLE	DISTURBED_UNABLE	AC_DISTURBED_UNABLE

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FC_mode	State	Ability	NewState	DGPMessage
FC_C	OCCUPIED_ABLE	UNABLE	DISTURBED_UNABLE	AC_DISTURBED_UNABLE
FC_U	OCCUPIED_UNABLE	ABLE	DISTURBED_ABLE	AC_DISTURBED_ABLE
FC_P	OCCUPIED_UNABLE	ABLE	DISTURBED_ABLE	AC_DISTURBED_ABLE
FC_U	OCCUPIED_UNABLE	UNABLE	DISTURBED_UNABLE	AC_DISTURBED_UNABLE
FC_P	OCCUPIED_UNABLE	UNABLE	DISTURBED_UNABLE	AC_DISTURBED_UNABLE

Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-AC:SSS:REQ: 2195	Axle Counter - Command reply - State transition with reporting with fault repair information
Given an Axle Counter element with an open FC command with <FC_mode> mode And the current element state is <State> And status message contains Occupancy Status OCCUPIED And status message contains Force Ability <Ability> When receiving the status message from TDS Then SEA shall set the current element state to <NewState> And SEA shall report the element state e_track_occupied_es with command reply flag true to CC And SEA shall send the fault notification sum_fault_rep with fault type e_if_error to CC And SEA shall report the diagnosis aspect message <DGPMessage> to DGP	

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FC_mode	State	Ability	NewState	DGPMessage
FC_U	DISTURBED_ABLE	ABLE	OCCUPIED_ABLE	AC_OCCUPIED_ABLE
FC_P	DISTURBED_ABLE	ABLE	OCCUPIED_ABLE	AC_OCCUPIED_ABLE
FC_C	DISTURBED_ABLE	ABLE	OCCUPIED_ABLE	AC_OCCUPIED_ABLE
FC_U	DISTURBED_ABLE	UNABLE	OCCUPIED_UNABLE	AC_OCCUPIED_UNABLE
FC_P	DISTURBED_ABLE	UNABLE	OCCUPIED_UNABLE	AC_OCCUPIED_UNABLE
FC_C	DISTURBED_ABLE	UNABLE	OCCUPIED_UNABLE	AC_OCCUPIED_UNABLE
FC_U	DISTURBED_UNABLE	UNABLE	OCCUPIED_UNABLE	AC_OCCUPIED_UNABLE
FC_P	DISTURBED_UNABLE	UNABLE	OCCUPIED_UNABLE	AC_OCCUPIED_UNABLE
FC_U	DISTURBED_UNABLE	ABLE	OCCUPIED_ABLE	AC_OCCUPIED_ABLE
FC_P	DISTURBED_UNABLE	ABLE	OCCUPIED_ABLE	AC_OCCUPIED_ABLE

Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-AC:SSS:REQ: 2148	Axle Counter - Command reply - Transition by sweeping train state
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Given an Axle Counter element with an open FC command with FC_P mode
And the current element state is <State>
And status message contains Occupancy Status <OccupStatus>
And status message contains Force Ability UNABLE

When receiving the status message from TDS

Then SEA shall set the current element state to <NewState>
And SEA shall report the diagnosis aspect message <DGPMMessage> to DGP

State	OccupStatus	NewState	DGPMMessage
OCCUPIED_ABLE	WAITING_FOR_SWEEPING_TRAIN	OCCUPIED_UNABLE	AC_OCCUPIED_UNABLE
OCCUPIED_ABLE	SWEEPING_TRAIN_DETECTED	OCCUPIED_UNABLE	AC_OCCUPIED_UNABLE
DISTURBED_ABLE	WAITING_FOR_SWEEPING_TRAIN	DISTURBED_UNABLE	AC_DISTURBED_UNABLE
DISTURBED_ABLE	SWEEPING_TRAIN_DETECTED	DISTURBED_UNABLE	AC_DISTURBED_UNABLE

Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-AC:SSS:REQ: 2149	Axle Counter - Command reply - Close open FC_P command with sweeping train state
Given an Axle Counter element with an open FC command with FC_P mode And the current element state is <State> And status message contains Occupancy Status <OccupStatus> And status message contains Force Ability UNABLE When receiving the status message from TDS	

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Then SEA shall close the open FC command with FC_P mode

State	OccupStatus
OCCUPIED_UNABLE	WAITING_FOR_SWEEPING_TRAIN
OCCUPIED_UNABLE	SWEEPING_TRAIN_DETECTED
DISTURBED_UNABLE	WAITING_FOR_SWEEPING_TRAIN
DISTURBED_UNABLE	SWEEPING_TRAIN_DETECTED

Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-AC:SSS:REQ: 2194 **Axle Counter - Command Reply - Close open FC_P command with repeated OCCUPIED / DISTURBED state**

Given an Axle Counter element with an open FC command with <FC_mode> mode

And the current element state is <State>

When receiving the status message with Occupancy Status <OccupStatus> and Force Ability <Ability> from TDS

Then SEA shall close the open FC command with FC_P mode

And SEA shall report the element state <CCState> with command reply flag true to CC

FC_mode	State	OccupStatus	Ability	CCState
FC_U	OCCUPIED_ABLE	OCCUPIED	ABLE	e_track_occupied_es
FC_P	OCCUPIED_ABLE	OCCUPIED	ABLE	e_track_occupied_es
FC_C	OCCUPIED_ABLE	OCCUPIED	ABLE	e_track_occupied_es

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FC_mode	State	OccupStatus	Ability	CCState
FC_U	OCCUPIED_UNABLE	OCCUPIED	UNABLE	e_track_occupied_es
FC_P	OCCUPIED_UNABLE	OCCUPIED	UNABLE	e_track_occupied_es
FC_U	DISTURBED_ABLE	DISTURBED	ABLE	e_track_not_regular_es
FC_P	DISTURBED_ABLE	DISTURBED	ABLE	e_track_not_regular_es
FC_C	DISTURBED_ABLE	DISTURBED	ABLE	e_track_not_regular_es
FC_U	DISTURBED_UNABLE	DISTURBED	UNABLE	e_track_not_regular_es
FC_P	DISTURBED_UNABLE	DISTURBED	UNABLE	e_track_not_regular_es

Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-AC:SSS:REQ: 2155	Axle Counter - Command reply - Ignore sweeping train as reply to FC_C / FC_U
Given an Axle Counter element with an open FC command with <FC_mode> mode And status message contains Occupancy Status <OccupStatus> When receiving the status message from TDS Then SEA shall report the plain text diagnosis message "TDS-IF-ERROR: Unexpected sweeping train Occupancy Status after <FC_mode> command for AC x" to DGP	

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FC_mode	OccupStatus
FC_C	WAITING_FOR_SWEEPING_TRAIN
FC_C	SWEEPING_TRAIN_DETECTED
FC_U	WAITING_FOR_SWEEPING_TRAIN
FC_U	SWEEPING_TRAIN_DETECTED

Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-AC:SSS:REQ: 2184	Axle Counter - Spontaneous message - Ignore sweeping train status message in UNDEFINED state								
Given the current element state is UNDEFINED And status message contains Occupancy Status <OccupStatus> And status message contains Force Ability <Ability> When receiving the status message from TDS Then SEA shall report the plain text diagnosis message "TDS-IF-ERROR: Unexpected sweeping train Occupancy Status without FC command for AC x" to DGP <table> <tr> <th>OccupStatus</th><th>Ability</th></tr> <tr> <td>WAITING_FOR_SWEEPING_TRAIN</td><td>UNABLE</td></tr> <tr> <td>WAITING_FOR_SWEEPING_TRAIN</td><td>ABLE</td></tr> <tr> <td>SWEEPING_TRAIN_DETECTED</td><td>UNABLE</td></tr> </table>		OccupStatus	Ability	WAITING_FOR_SWEEPING_TRAIN	UNABLE	WAITING_FOR_SWEEPING_TRAIN	ABLE	SWEEPING_TRAIN_DETECTED	UNABLE
OccupStatus	Ability								
WAITING_FOR_SWEEPING_TRAIN	UNABLE								
WAITING_FOR_SWEEPING_TRAIN	ABLE								
SWEEPING_TRAIN_DETECTED	UNABLE								

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OccupStatus		Ability	
SWEEPING_ TRAIN_ DETECTED		ABLE	
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-AC:SSS:REQ: 2185	Axle Counter - Spontaneous message - Ignore sweeping train state with no open command
Given an Axle Counter element with no open reset command And the current element state is <State> And status message contains Occupancy Status <OccupStatus> When receiving the status message from TDS Then SEA shall report the plain text diagnosis message "TDS-IF-ERROR: Unexpected sweeping train Occupancy Status without FC command for AC x" to DGP	
State	OccupStatus
OCCUPIED_ ABLE	WAITING_ FOR_ SWEEPING_ TRAIN
OCCUPIED_ ABLE	SWEEPING_ TRAIN_ DETECTED
OCCUPIED_ UNABLE	WAITING_ FOR_ SWEEPING_ TRAIN
OCCUPIED_ UNABLE	SWEEPING_ TRAIN_ DETECTED
OCCUPIED_ TIMED	WAITING_ FOR_ SWEEPING_ TRAIN
OCCUPIED_ TIMED	SWEEPING_ TRAIN_ DETECTED
DISTURBED_ ABLE	WAITING_ FOR_ SWEEPING_ TRAIN

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State	OccupStatus
DISTURBED_ ABLE	SWEEPING_ TRAIN_ DETECTED
DISTURBED_ UNABLE	WAITING_ FOR_ SWEEPING_ TRAIN
DISTURBED_ UNABLE	SWEEPING_ TRAIN_ DETECTED
DISTURBED_ TIMED	WAITING_ FOR_ SWEEPING_ TRAIN
DISTURBED_ TIMED	SWEEPING_ TRAIN_ DETECTED
FREE	WAITING_ FOR_ SWEEPING_ TRAIN
FREE	SWEEPING_ TRAIN_ DETECTED

Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-AC:SSS:REQ: 2157	Axle Counter - Spontaneous message - Ignore message Additional Information		
When receiving the Additional Information message with Wheel Diameter value within valid range and Speed value within valid range from TDS Then SEA shall report the plain text diagnosis message "TDS-IF-ERROR: Unexpected Additional Information message for AC x" to DGP			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit

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Comments:	According to SCI-TDS the Wheel Diameter is within 0x0001 and 0x9999 and the Speed is within 0x0000 and 0x9999.
Satisfies:	

EC-MUX-AC:SSS:REQ: 2158	Axle Counter - Spontaneous message - Disconnect on message TDP Status																						
When receiving the TDP Status message with State of Passing <StateOfPassing> and Direction of Passing <DirectionOfPassing> from TDS Then SEA shall report the plain text diagnosis message "TDS-IF-ERROR: Unexpected TDP Status message for AC x" to DGP And SEA shall execute EC-MUX safety reaction OC Disconnection And SEA shall report the Axle Counter no_connection diagnosis device fault message to DGP																							
<table><tr><th>StateOfPassing</th><th>DirectionOfPassing</th></tr><tr><td>NOT_ PASSED</td><td>REFERENCE_ DIRECTION</td></tr><tr><td>NOT_ PASSED</td><td>AGAINST_ REFERENCE_ DIRECTION</td></tr><tr><td>NOT_ PASSED</td><td>WITHOUT_ INDICATED_ DIRECTION</td></tr><tr><td>PASSED</td><td>AGAINST_ REFERENCE_ DIRECTION</td></tr><tr><td>PASSED</td><td>REFERENCE_ DIRECTION</td></tr><tr><td>PASSED</td><td>WITHOUT_ INDICATED_ DIRECTION</td></tr><tr><td>DISTURBED</td><td>AGAINST_ REFERENCE_ DIRECTION</td></tr><tr><td>DISTURBED</td><td>REFERENCE_ DIRECTION</td></tr><tr><td>DISTURBED</td><td>WITHOUT_ INDICATED_ DIRECTION</td></tr></table>				StateOfPassing	DirectionOfPassing	NOT_ PASSED	REFERENCE_ DIRECTION	NOT_ PASSED	AGAINST_ REFERENCE_ DIRECTION	NOT_ PASSED	WITHOUT_ INDICATED_ DIRECTION	PASSED	AGAINST_ REFERENCE_ DIRECTION	PASSED	REFERENCE_ DIRECTION	PASSED	WITHOUT_ INDICATED_ DIRECTION	DISTURBED	AGAINST_ REFERENCE_ DIRECTION	DISTURBED	REFERENCE_ DIRECTION	DISTURBED	WITHOUT_ INDICATED_ DIRECTION
StateOfPassing	DirectionOfPassing																						
NOT_ PASSED	REFERENCE_ DIRECTION																						
NOT_ PASSED	AGAINST_ REFERENCE_ DIRECTION																						
NOT_ PASSED	WITHOUT_ INDICATED_ DIRECTION																						
PASSED	AGAINST_ REFERENCE_ DIRECTION																						
PASSED	REFERENCE_ DIRECTION																						
PASSED	WITHOUT_ INDICATED_ DIRECTION																						
DISTURBED	AGAINST_ REFERENCE_ DIRECTION																						
DISTURBED	REFERENCE_ DIRECTION																						
DISTURBED	WITHOUT_ INDICATED_ DIRECTION																						
<table><tr><td>Type:</td><td colspan="3">FUR - Functional Requirement</td></tr><tr><td>Status:</td><td>draft</td><td>Stability:</td><td>stable</td></tr><tr><td>Target Release:</td><td>SEA SW GEN 3.0</td><td>Priority:</td><td>mandatory</td></tr><tr><td>Originator:</td><td>Technics (R&D)</td><td>Allocation:</td><td>SW EC-MUX</td></tr><tr><td>Verification Method:</td><td>Test</td><td>Verification Step:</td><td>module test, subit</td></tr></table>				Type:	FUR - Functional Requirement			Status:	draft	Stability:	stable	Target Release:	SEA SW GEN 3.0	Priority:	mandatory	Originator:	Technics (R&D)	Allocation:	SW EC-MUX	Verification Method:	Test	Verification Step:	module test, subit
Type:	FUR - Functional Requirement																						
Status:	draft	Stability:	stable																				
Target Release:	SEA SW GEN 3.0	Priority:	mandatory																				
Originator:	Technics (R&D)	Allocation:	SW EC-MUX																				
Verification Method:	Test	Verification Step:	module test, subit																				

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Comments:	According to the SCI-TDS the TDP Status message shall never be addressed to the AC element, therefore the OC of this AC element will be disconnected. TDP Status message to be addressed to the TDP element only.
Satisfies:	

3.1.1.3 Operational - Axle Counter - Irregularities

In order to control the element state of the Axle Counter

As a SEA

I want to be able to receive state messages from TDS and evaluate the consistency of its attributes before executing the internal element state transition and reporting the state to CC.

For requirement language see Chapter 2.6 "Requirement Language".

The Disturbance Status does not influence the state change.

If a status message contains the Occupancy Status "VACANT" or "OCCUPIED" the consistent value for the Disturbance Status is "NOT APPLICABLE". Values "TECHNICAL" / "OPERATIONAL" to be considered as inconsistent.

If a status message contains the Occupancy Status "DISTURBED" the consistent value for the Disturbance Status is "TECHNICAL" / "OPERATIONAL". Value "NOT APPLICABLE" to be considered as inconsistent.

A consistent value for the Disturbance Status doesn't trigger an error message.

An inconsistent value for the Disturbance Status triggers an error message.

The Disturbance Status is applicable for both cases where we have an open command and where we don't have an open command.

According to SCI-TDS the Filling Level can have a value between 0x0001 and 0xFFFE only in the reply to the command Update Filling Level, which is not supported by the CC interface. For this reason "NOT_APPLICABLE" is the only consistent value.

The attribute POM Status is intended to be used for the Track Circuit element only. For the Axle Counter element "NOT_APPLICABLE" is the only consistent value.

The SSS-TDS does not define clearly which states will be regularly notified with which Change Trigger. For this reason SEA shall handle the Change Trigger attribute quite tolerantly and accept most of the possible combinations, except those obviously inconsistent (e.g. Occupancy State "VACANT" / "OCCUPIED" reported with the Change Trigger "TECHNICAL_FAILURE").

All requirements in this chapter are also applicable for the messages received during the EC-MUX - TDS connection establishment.

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Given SEA with a configured Axle Counter element

And the mapping between SEA diagnosis aspect and DGP diagnosis value is defined

SEA diagnosis aspect	DGP diagnosis value
AC_FREE	0
AC_OCCUPIED_ABLE	1
AC_OCCUPIED_UNABLE	2
AC_OCCUPIED_TIMED	3
AC_DISTURBED_ABLE	4
AC_DISTURBED_UNABLE	5
AC_DISTURBED_TIMED	6
AC_UNDEFINED	7

EC-MUX-AC:SSS:REQ: 2156	Axle Counter - Consistent Disturbance Status without error reporting in disturbed states					
When receiving the status message with Occupancy Status DISTURBED and Disturbance Status <DisturbanceStatus> from TDS Then SEA shall process the message <table><tr><td>DisturbanceStatus</td></tr><tr><td>TECHNICAL</td></tr><tr><td>OPERATIONAL</td></tr></table>				DisturbanceStatus	TECHNICAL	OPERATIONAL
DisturbanceStatus						
TECHNICAL						
OPERATIONAL						
Type:	FUR - Functional Requirement					
Status:	draft	Stability:	stable			
Target Release:	SEA SW GEN 3.0	Priority:	mandatory			
Originator:	Technics (R&D)	Allocation:	SW EC-MUX			
Verification Method:	Test	Verification Step:	module test, subit			
Comments:						
Satisfies:						

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EC-MUX-AC:SSS:REQ: 2162	Axle Counter - Consistent Disturbance Status without error reporting in regular states					
When receiving the status message with Occupancy Status <OccupStatus> and Disturbance Status NOT_APPLICABLE from TDS Then SEA shall process the message <table><tr><td>OccupStatus</td></tr><tr><td>VACANT</td></tr><tr><td>OCCUPIED</td></tr></table>				OccupStatus	VACANT	OCCUPIED
OccupStatus						
VACANT						
OCCUPIED						
Type:	FUR - Functional Requirement					
Status:	draft	Stability:	stable			
Target Release:	SEA SW GEN 3.0	Priority:	mandatory			
Originator:	Technics (R&D)	Allocation:	SW EC-MUX			
Verification Method:	Test	Verification Step:	module test, subit			
Comments:						
Satisfies:						

EC-MUX-AC:SSS:REQ: 2163	Axle Counter - Disconnect on Inconsistent Disturbance Status in disturbed states		
When receiving the status message with Occupancy Status DISTURBED and Disturbance Status NOT_APPLICABLE from TDS Then SEA shall report the plain text diagnosis message "TDS-IF-ERROR: Inconsistent Disturbance Status for AC x" to DGP And SEA shall execute EC-MUX safety reaction OC Disconnection And SEA shall report the Axle Counter no_connection diagnosis device fault message to DGP			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX

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Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-AC:SSS:REQ: 2164	Axle Counter - Disconnect on Inconsistent Disturbance Status in regular states												
When receiving the status message with Occupancy Status <OccupStatus> and Disturbance Status <DisturbanceStatus> from TDS Then SEA shall report the plain text diagnosis message "TDS-IF-ERROR: Inconsistent Disturbance Status for AC x" to DGP And SEA shall execute EC-MUX safety reaction OC Disconnection And SEA shall report the Axle Counter no_connection diagnosis device fault message to DGP <table><tr><th>OccupStatus</th><th>DisturbanceStatus</th></tr><tr><td>VACANT</td><td>TECHNICAL</td></tr><tr><td>OCCUPIED</td><td>OPERATIONAL</td></tr><tr><td>VACANT</td><td>OPERATIONAL</td></tr><tr><td>OCCUPIED</td><td>TECHNICAL</td></tr></table>				OccupStatus	DisturbanceStatus	VACANT	TECHNICAL	OCCUPIED	OPERATIONAL	VACANT	OPERATIONAL	OCCUPIED	TECHNICAL
OccupStatus	DisturbanceStatus												
VACANT	TECHNICAL												
OCCUPIED	OPERATIONAL												
VACANT	OPERATIONAL												
OCCUPIED	TECHNICAL												
Type:	FUR - Functional Requirement												
Status:	draft	Stability:	stable										
Target Release:	SEA SW GEN 3.0	Priority:	mandatory										
Originator:	Technics (R&D)	Allocation:	SW EC-MUX										
Verification Method:	Test	Verification Step:	module test, subit										
Comments:													
Satisfies:													

EC-MUX-AC:SSS:REQ: 2165	Axle Counter - Inconsistent POM Status with error reporting
When receiving the status message with POM Status <POM_Status> from TDS	

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Then SEA shall process the message And SEA shall report the plain text diagnosis message "TDS-IF-ERROR: Inconsistent POM Status for AC x" to DGP															
<table><tr><td colspan="4">POM_Status</td></tr><tr><td colspan="4">POWER_SUPPLY_IS_OK</td></tr><tr><td colspan="4">POWER_SUPPLY_IS_NOK</td></tr></table>				POM_Status				POWER_SUPPLY_IS_OK				POWER_SUPPLY_IS_NOK			
POM_Status															
POWER_SUPPLY_IS_OK															
POWER_SUPPLY_IS_NOK															
Type:	FUR - Functional Requirement														
Status:	draft	Stability:	stable												
Target Release:	SEA SW GEN 3.0	Priority:	mandatory												
Originator:	Technics (R&D)	Allocation:	SW EC-MUX												
Verification Method:	Test	Verification Step:	module test, subit												
Comments:															
Satisfies:															

EC-MUX-AC:SSS:REQ: 2166	Axle Counter - Consistent POM Status without error reporting		
When receiving the status message with POM Status NOT_APPLICABLE from TDS			
Then SEA shall process the message			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-AC:SSS:REQ:	Axle Counter - Inconsistent Filling Level with error reporting
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When receiving the status message with Filling Level different from NOT_APPLICABLE from TDS Then SEA shall process the message And SEA shall report the plain text diagnosis message "TDS-IF-ERROR: Inconsistent Filling Level for AC x" to DGP			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-AC:SSS:REQ: 2168	Axle Counter - Consistent Filling Level without error reporting		
When receiving the status message with Filling Level NOT_APPLICABLE from TDS			
Then SEA shall change the element state as included in status message			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-AC:SSS:REQ: 2169	Axle Counter - Consistent Change Trigger without error reporting		
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When receiving the status message with Occupancy Status <OccupStatus> and Change Trigger <ChangeTrigger> from TDS

Then SEA shall process the message

OccupStatus	ChangeTrigger
VACANT	PASSING_ DETECTED
VACANT	COMMAND_ FROM_ EIL
VACANT	COMMAND_ FROM_ MAINTAINER
VACANT	INITIAL_ SECTION_ STATE
VACANT	INTERNAL_ TRIGGER
VACANT	NOT_ APPLICABLE
OCCUPIED	PASSING_ DETECTED
OCCUPIED	COMMAND_ FROM_ EIL
OCCUPIED	COMMAND_ FROM_ MAINTAINER
OCCUPIED	INITIAL_ SECTION_ STATE
OCCUPIED	INTERNAL_ TRIGGER
OCCUPIED	NOT_ APPLICABLE
WAITING_ FOR_ SWEEPING_ TRAIN	PASSING_ DETECTED
WAITING_ FOR_ SWEEPING_ TRAIN	COMMAND_ FROM_ EIL
WAITING_ FOR_ SWEEPING_ TRAIN	COMMAND_ FROM_ MAINTAINER
WAITING_ FOR_ SWEEPING_ TRAIN	INTERNAL_ TRIGGER
WAITING_ FOR_ SWEEPING_ TRAIN	NOT_ APPLICABLE
SWEEPING_ TRAIN_ DETECTED	PASSING_ DETECTED
SWEEPING_ TRAIN_ DETECTED	COMMAND_ FROM_ EIL
SWEEPING_ TRAIN_ DETECTED	COMMAND_ FROM_ MAINTAINER
SWEEPING_ TRAIN_ DETECTED	INTERNAL_ TRIGGER
SWEEPING_ TRAIN_ DETECTED	NOT_ APPLICABLE

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Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-AC:SSS:REQ: 2170	Axle Counter - Disconnect on Inconsistent Change Trigger in regular states																
When receiving the status message with Occupancy Status <OccupStatus> and Change Trigger <ChangeTrigger> from TDS Then SEA shall report the plain text diagnosis message "TDS-IF-ERROR: Inconsistent Change Trigger for AC x" to DGP And SEA shall execute EC-MUX safety reaction OC Disconnection And SEA shall report the Axle Counter no_connection diagnosis device fault message to DGP <table><tr><th>OccupStatus</th><th>ChangeTrigger</th></tr><tr><td>VACANT</td><td>TECHNICAL_ FAILURE</td></tr><tr><td>OCCUPIED</td><td>TECHNICAL_ FAILURE</td></tr><tr><td>WAITING_ FOR_ SWEEPING_ TRAIN</td><td>TECHNICAL_ FAILURE</td></tr><tr><td>WAITING_ FOR_ SWEEPING_ TRAIN</td><td>INITIAL_ SECTION_ STATE</td></tr><tr><td>SWEEPING_ TRAIN_ DETECTED</td><td>TECHNICAL_ FAILURE</td></tr><tr><td>SWEEPING_ TRAIN_ DETECTED</td><td>INITIAL_ SECTION_ STATE</td></tr></table>				OccupStatus	ChangeTrigger	VACANT	TECHNICAL_ FAILURE	OCCUPIED	TECHNICAL_ FAILURE	WAITING_ FOR_ SWEEPING_ TRAIN	TECHNICAL_ FAILURE	WAITING_ FOR_ SWEEPING_ TRAIN	INITIAL_ SECTION_ STATE	SWEEPING_ TRAIN_ DETECTED	TECHNICAL_ FAILURE	SWEEPING_ TRAIN_ DETECTED	INITIAL_ SECTION_ STATE
OccupStatus	ChangeTrigger																
VACANT	TECHNICAL_ FAILURE																
OCCUPIED	TECHNICAL_ FAILURE																
WAITING_ FOR_ SWEEPING_ TRAIN	TECHNICAL_ FAILURE																
WAITING_ FOR_ SWEEPING_ TRAIN	INITIAL_ SECTION_ STATE																
SWEEPING_ TRAIN_ DETECTED	TECHNICAL_ FAILURE																
SWEEPING_ TRAIN_ DETECTED	INITIAL_ SECTION_ STATE																
Type:	FUR - Functional Requirement																
Status:	draft	Stability:	stable														
Target Release:	SEA SW GEN 3.0	Priority:	mandatory														
Originator:	Technics (R&D)	Allocation:	SW EC-MUX														
Verification Method:	Test	Verification Step:	module test, subit														

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Comments:	
Satisfies:	

EC-MUX-AC:SSS:REQ: 2171	Axle Counter - Consistent Change Trigger without error reporting in disturbed states										
When receiving the status message with Occupancy Status DISTURBED and Change Trigger <ChangeTrigger> from TDS Then SEA shall process the message <table><tr><td>ChangeTrigger</td></tr><tr><td>PASSING_ DETECTED</td></tr><tr><td>COMMAND_ FROM_ EIL</td></tr><tr><td>COMMAND_ FROM_ MAINTAINER</td></tr><tr><td>INITIAL_ SECTION_ STATE</td></tr><tr><td>TECHNICAL_ FAILURE</td></tr><tr><td>INTERNAL_ TRIGGER</td></tr><tr><td>NOT_ APPLICABLE</td></tr></table>				ChangeTrigger	PASSING_ DETECTED	COMMAND_ FROM_ EIL	COMMAND_ FROM_ MAINTAINER	INITIAL_ SECTION_ STATE	TECHNICAL_ FAILURE	INTERNAL_ TRIGGER	NOT_ APPLICABLE
ChangeTrigger											
PASSING_ DETECTED											
COMMAND_ FROM_ EIL											
COMMAND_ FROM_ MAINTAINER											
INITIAL_ SECTION_ STATE											
TECHNICAL_ FAILURE											
INTERNAL_ TRIGGER											
NOT_ APPLICABLE											
Type:	FUR - Functional Requirement										
Status:	draft	Stability:	stable								
Target Release:	SEA SW GEN 3.0	Priority:	mandatory								
Originator:	Technics (R&D)	Allocation:	SW EC-MUX								
Verification Method:	Test	Verification Step:	module test, subit								
Comments:	In a disturbed state every possible change trigger will be accepted and is considered consistent.										
Satisfies:											

EC-MUX-AC:SSS:REQ: 2172	Axle Counter - Inconsistent Ability in VACANT message with error reporting
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When receiving the status message with Occupancy Status VACANT and Force Ability ABLE from TDS

Then SEA shall process the message

And SEA shall report the plain text diagnosis message "TDS-IF-ERROR: Inconsistent Ability to be forced with VACANT message for AC x" to DGP

Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-AC:SSS:REQ: 2191 Axle Counter - Consistent Ability in VACANT message

When receiving the status message with Occupancy Status VACANT and Force Ability UNABLE from TDS

Then SEA shall process the message

Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-AC:SSS:REQ: 2173 Axle Counter - Inconsistent Ability in sweeping train message with error reporting

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When receiving the status message with Occupancy Status <OccupStatus> and Force Ability ABLE from TDS

Then SEA shall process the message

And SEA shall report the plain text diagnosis message "TDS-IF-ERROR: Inconsistent Ability to be forced with sweeping train message for AC x" to DGP

OccupStatus
WAITING_ FOR_ SWEEPING_ TRAIN
SWEEPING_ TRAIN_ DETECTED

Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-AC:SSS:REQ: 2192	Axle Counter - Consistent Ability in sweeping train message					
When receiving the status message with Occupancy Status <OccupStatus> and Force Ability UNABLE from TDS Then SEA shall process the message <table><tr><th>OccupStatus</th></tr><tr><td>WAITING_ FOR_ SWEEPING_ TRAIN</td></tr><tr><td>SWEEPING_ TRAIN_ DETECTED</td></tr></table>				OccupStatus	WAITING_ FOR_ SWEEPING_ TRAIN	SWEEPING_ TRAIN_ DETECTED
OccupStatus						
WAITING_ FOR_ SWEEPING_ TRAIN						
SWEEPING_ TRAIN_ DETECTED						
Type:	FUR - Functional Requirement					
Status:	draft	Stability:	stable			
Target Release:	SEA SW GEN 3.0	Priority:	mandatory			

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Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-AC:SSS:REQ: 2193	Axle Counter - Consistent sweeping train status message with any Disturbance Status																
When receiving the status message with Occupancy Status <OccupStatus> and Disturbance Status <DisturbanceStatus> from TDS Then SEA shall process the message <table><tr><th>OccupStatus</th><th>DisturbanceStatus</th></tr><tr><td>WAITING_ FOR_ SWEEPING_ TRAIN</td><td>OPERATIONAL</td></tr><tr><td>WAITING_ FOR_ SWEEPING_ TRAIN</td><td>TECHNICAL</td></tr><tr><td>WAITING_ FOR_ SWEEPING_ TRAIN</td><td>NOT_ APPLICABLE</td></tr><tr><td>SWEEPING_ TRAIN_ DETECTED</td><td>OPERATIONAL</td></tr><tr><td>SWEEPING_ TRAIN_ DETECTED</td><td>TECHNICAL</td></tr><tr><td>SWEEPING_ TRAIN_ DETECTED</td><td>NOT_ APPLICABLE</td></tr></table>				OccupStatus	DisturbanceStatus	WAITING_ FOR_ SWEEPING_ TRAIN	OPERATIONAL	WAITING_ FOR_ SWEEPING_ TRAIN	TECHNICAL	WAITING_ FOR_ SWEEPING_ TRAIN	NOT_ APPLICABLE	SWEEPING_ TRAIN_ DETECTED	OPERATIONAL	SWEEPING_ TRAIN_ DETECTED	TECHNICAL	SWEEPING_ TRAIN_ DETECTED	NOT_ APPLICABLE
OccupStatus	DisturbanceStatus																
WAITING_ FOR_ SWEEPING_ TRAIN	OPERATIONAL																
WAITING_ FOR_ SWEEPING_ TRAIN	TECHNICAL																
WAITING_ FOR_ SWEEPING_ TRAIN	NOT_ APPLICABLE																
SWEEPING_ TRAIN_ DETECTED	OPERATIONAL																
SWEEPING_ TRAIN_ DETECTED	TECHNICAL																
SWEEPING_ TRAIN_ DETECTED	NOT_ APPLICABLE																
Type:	FUR - Functional Requirement																
Status:	draft	Stability:	stable														
Target Release:	SEA SW GEN 3.0	Priority:	mandatory														
Originator:	Technics (R&D)	Allocation:	SW EC-MUX														
Verification Method:	Test	Verification Step:	module test, subit														
Comments:																	
Satisfies:																	

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3.1.2 Technical - Axle Counter

3.1.2.1 Connection - Axle Counter

In order to supervise the Axle Counter

As a SEA

I want to be able to supervise connections and when changes occur then

- maintain the element state
- report its changes to CC
- report the connection state changes to CC and DGP

For requirement language see Chapter 2.6 "Requirement Language".

Given SEA with a configured Axle Counter element

And the mapping between SEA diagnosis aspect and DGP diagnosis value is defined

SEA diagnosis aspect	DGP diagnosis value
AC_ FREE	0
AC_ OCCUPIED_ ABLE	1
AC_ OCCUPIED_ UNABLE	2
AC_ OCCUPIED_ TIMED	3
AC_ DISTURBED_ ABLE	4
AC_ DISTURBED_ UNABLE	5
AC_ DISTURBED_ TIMED	6
AC_ UNDEFINED	7

EC-MUX-AC:SSS:REQ: 2174	Axle Counter - [Re-]Connecting to CC - TDS CONNECTED - General inquiry
Given the current element state is <State> And CC connection state is NOT_CONNECTED And EC-MUX - TDS connection state is CONNECTED When establishing the connection to CC Then SEA shall report the element state <CCState> with command reply flag false to CC And SEA shall send the fault notification <FaultMsg> with fault type e_if_error to CC	

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State	CCState	FaultMsg
OCCUPIED_ ABLE	e_track_occupied_es	sum_fault_rep
OCCUPIED_ UNABLE	e_track_occupied_es	sum_fault_rep
OCCUPIED_ TIMED	e_track_occupied_es	sum_fault_rep
FREE	e_track_free_es	sum_fault_rep
DISTURBED_ ABLE	e_track_not_regular_es	sum_fault
DISTURBED_ UNABLE	e_track_not_regular_es	sum_fault
DISTURBED_ TIMED	e_track_not_regular_es	sum_fault
UNDEFINED	e_undefined_es	sum_fault

Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-AC:SSS:REQ: 2175	Axle Counter - [Re-]Connecting to CC - TDS NOT_CONNECTED - General inquiry
Given the current element state is UNDEFINED And CC connection state is NOT_CONNECTED And EC-MUX - TDS connection state is NOT_CONNECTED When establishing the connection to CC Then SEA shall report the element state e_undefined_es with command reply flag false to CC And SEA shall send the fault notification sum_fault with fault type e_if_error to CC	

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Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-AC:SSS:REQ: 2176	Axle Counter - Connection loss to CC											
Given the current element state is <State> And CC connection state is CONNECTED When losing the connection to CC Then SEA shall do nothing <table><tr><th>State</th></tr><tr><td>OCCUPIED_ ABLE</td></tr><tr><td>OCCUPIED_ UNABLE</td></tr><tr><td>OCCUPIED_ TIMED</td></tr><tr><td>DISTURBED_ ABLE</td></tr><tr><td>DISTURBED_ UNABLE</td></tr><tr><td>DISTURBED_ TIMED</td></tr><tr><td>FREE</td></tr><tr><td>UNDEFINED</td></tr></table>				State	OCCUPIED_ ABLE	OCCUPIED_ UNABLE	OCCUPIED_ TIMED	DISTURBED_ ABLE	DISTURBED_ UNABLE	DISTURBED_ TIMED	FREE	UNDEFINED
State												
OCCUPIED_ ABLE												
OCCUPIED_ UNABLE												
OCCUPIED_ TIMED												
DISTURBED_ ABLE												
DISTURBED_ UNABLE												
DISTURBED_ TIMED												
FREE												
UNDEFINED												
Type:	FUR - Functional Requirement											
Status:	draft	Stability:	stable									
Target Release:	SEA SW GEN 3.0	Priority:	mandatory									
Originator:	Technics (R&D)	Allocation:	SW EC-MUX									
Verification Method:	Test	Verification Step:	module test, subit									

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Comments:	
Satisfies:	

EC-MUX-AC:SSS:REQ: 2179	Axle Counter - Connection loss between EC-MUX and TDS in regular states							
Given the current element state is <State> And CC connection state is CONNECTED And EC-MUX - TDS connection state is CONNECTED When losing the connection between EC-MUX and TDS Then SEA shall set the current element state to UNDEFINED And SEA shall report the element state e_undefined_es with command reply flag false to CC And SEA shall send the fault notification sum_fault with fault type e_if_error to CC And SEA shall report the aspect diagnosis message AC_UNDEFINED to DGP And SEA shall report the Axle Counter no_connection diagnosis device fault message to DGP <table><tr><th>State</th></tr><tr><td>OCCUPIED_ ABLE</td></tr><tr><td>OCCUPIED_ UNABLE</td></tr><tr><td>OCCUPIED_ TIMED</td></tr><tr><td>FREE</td></tr></table>				State	OCCUPIED_ ABLE	OCCUPIED_ UNABLE	OCCUPIED_ TIMED	FREE
State								
OCCUPIED_ ABLE								
OCCUPIED_ UNABLE								
OCCUPIED_ TIMED								
FREE								
Type:	FUR - Functional Requirement							
Status:	draft	Stability:	stable					
Target Release:	SEA SW GEN 3.0	Priority:	mandatory					
Originator:	Technics (R&D)	Allocation:	SW EC-MUX					
Verification Method:	Test	Verification Step:	module test, subit					
Comments:								
Satisfies:								

EC-MUX-AC:SSS:REQ:	Axle Counter - Connection loss between EC-MUX and TDS in disturbed
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2180	states						
Given the current element state is <State> And CC connection state is CONNECTED And EC-MUX - TDS connection state is CONNECTED When losing the connection between EC-MUX and TDS Then SEA shall set the current element state to UNDEFINED And SEA shall report the element state e_undefined_es with command reply flag false to CC And SEA shall report the aspect diagnosis message AC_UNDEFINED to DGP And SEA shall report the Axle Counter no_connection diagnosis device fault message to DGP <table><tr><th>State</th></tr><tr><td>DISTURBED_ ABLE</td></tr><tr><td>DISTURBED_ UNABLE</td></tr><tr><td>DISTURBED_ TIMED</td></tr></table>				State	DISTURBED_ ABLE	DISTURBED_ UNABLE	DISTURBED_ TIMED
State							
DISTURBED_ ABLE							
DISTURBED_ UNABLE							
DISTURBED_ TIMED							
Type:	FUR - Functional Requirement						
Status:	draft	Stability:	stable				
Target Release:	SEA SW GEN 3.0	Priority:	mandatory				
Originator:	Technics (R&D)	Allocation:	SW EC-MUX				
Verification Method:	Test	Verification Step:	module test, subit				
Comments:							
Satisfies:							

EC-MUX-AC:SSS:REQ: 2181	Axle Counter - Connection loss between EC-MUX and TDS - CC not connected
Given the current element state is <State> And CC connection state is NOT_CONNECTED And EC-MUX - TDS connection state is CONNECTED When losing the connection between EC-MUX and TDS Then SEA shall set the current element state to UNDEFINED And SEA shall report the aspect diagnosis message AC_UNDEFINED to DGP And SEA shall report the Axle Counter no_connection diagnosis device fault message to DGP	

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<table><tr><th>State</th></tr><tr><td>DISTURBED_ ABLE</td></tr><tr><td>DISTURBED_ UNABLE</td></tr><tr><td>DISTURBED_ TIMED</td></tr><tr><td>OCCUPIED_ ABLE</td></tr><tr><td>OCCUPIED_ UNABLE</td></tr><tr><td>OCCUPIED_ TIMED</td></tr><tr><td>FREE</td></tr></table>				State	DISTURBED_ ABLE	DISTURBED_ UNABLE	DISTURBED_ TIMED	OCCUPIED_ ABLE	OCCUPIED_ UNABLE	OCCUPIED_ TIMED	FREE
State											
DISTURBED_ ABLE											
DISTURBED_ UNABLE											
DISTURBED_ TIMED											
OCCUPIED_ ABLE											
OCCUPIED_ UNABLE											
OCCUPIED_ TIMED											
FREE											
Type:	FUR - Functional Requirement										
Status:	draft	Stability:	stable								
Target Release:	SEA SW GEN 3.0	Priority:	mandatory								
Originator:	Technics (R&D)	Allocation:	SW EC-MUX								
Verification Method:	Test	Verification Step:	module test, subit								
Comments:											
Satisfies:											

EC-MUX-AC:SSS:REQ: 2177	Axle Counter - Connection loss between EC-MUX and TDS - UNDEFINED				
Given the current state is UNDEFINED And TDS Occupancy Status is <OccupStatus> And TDS Force Ability is <Ability> And CC connection state is CONNECTED And EC-MUX - TDS connection state is CONNECTED When losing the connection between EC-MUX and TDS Then SEA shall report the Axle Counter no_connection diagnosis device fault message to DGP <table><tr><th>OccupStatus</th><th>Ability</th></tr><tr><td>WAITING_ FOR_ SWEEPING_ TRAIN</td><td>UNABLE</td></tr></table>		OccupStatus	Ability	WAITING_ FOR_ SWEEPING_ TRAIN	UNABLE
OccupStatus	Ability				
WAITING_ FOR_ SWEEPING_ TRAIN	UNABLE				

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OccupStatus		Ability	
WAITING_ FOR_ SWEEPING_ TRAIN		ABLE	
SWEEPING_ TRAIN_ DETECTED		UNABLE	
SWEEPING_ TRAIN_ DETECTED		ABLE	

Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-AC:SSS:REQ: 2178

Axle Counter - [Re-]Connecting between EC-MUX and TDS - TDS in VACANT / OCCUPIED state

Given the current element state is UNDEFINED
And EC-MUX - TDS connection state is NOT_CONNECTED
And TDS Occupancy Status is <OccupStatus>
And TDS Force Ability is <Ability>

When establishing the connection between EC-MUX and TDS

Then SEA shall set the current element state to <State>
And SEA shall report the element state <CCState> with command reply flag false to CC
And SEA shall send the fault notification sum_fault_rep with fault type e_if_error to CC
And SEA shall report the diagnosis aspect message <DGPMMessage> to DGP
And SEA shall report the Axle Counter connection_ok diagnosis device fault repair message to DGP

OccupStatus	Ability	State	CCState	DGPMMessage
VACANT	UNABLE	FREE	e_track_free_es	AC_FREE
OCCUPIED	ABLE	OCCUPIED_ ABLE	e_track_occupied_es	AC_OCCUPIED_ ABLE
OCCUPIED	UNABLE	OCCUPIED_ UNABLE	e_track_occupied_es	AC_OCCUPIED_ UNABLE

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Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-AC:SSS:REQ: 2198	Axle Counter - [Re-]Connecting between EC-MUX and TDS - TDS in DISTURBED state											
Given the current element state is UNDEFINED And EC-MUX - TDS connection state is NOT_CONNECTED And TDS Occupancy Status is DISTURBED And TDS Force Ability is <Ability> When establishing the connection between EC-MUX and TDS Then SEA shall set the current element state to <State> And SEA shall report the element state e_track_not_regular_es with command reply flag false to CC And SEA shall report the diagnosis aspect message <DGPMMessage> to DGP And SEA shall report the Axle Counter connection_ok diagnosis device fault repair message to DGP <table><tr><th>Ability</th><th>State</th><th>DGPMMessage</th></tr><tr><td>ABLE</td><td>DISTURBED_ ABLE</td><td>AC_ DISTURBED_ ABLE</td></tr><tr><td>UNABLE</td><td>DISTURBED_ UNABLE</td><td>AC_ DISTURBED_ UNABLE</td></tr></table>				Ability	State	DGPMMessage	ABLE	DISTURBED_ ABLE	AC_ DISTURBED_ ABLE	UNABLE	DISTURBED_ UNABLE	AC_ DISTURBED_ UNABLE
Ability	State	DGPMMessage										
ABLE	DISTURBED_ ABLE	AC_ DISTURBED_ ABLE										
UNABLE	DISTURBED_ UNABLE	AC_ DISTURBED_ UNABLE										
Type:	FUR - Functional Requirement											
Status:	draft	Stability:	stable									
Target Release:	SEA SW GEN 3.0	Priority:	mandatory									
Originator:	Technics (R&D)	Allocation:	SW EC-MUX									
Verification Method:	Test	Verification Step:	module test, subit									
Comments:												
Satisfies:												

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EC-MUX-AC:SSS:REQ: 2197	Axle Counter - [Re-]Connecting between EC-MUX and TDS - TDS in sweeping train state					
Given the current element state is UNDEFINED And EC-MUX - TDS connection state is NOT_CONNECTED And TDS Occupancy Status is <OccupStatus> And TDS Force Ability is UNABLE When establishing the connection between EC-MUX and TDS Then SEA shall report the plain text diagnosis message "TDS-IF-ERROR: Invalid sequence - sweeping train as initial Occupancy Status for AC x" to DGP And SEA shall report the Axle Counter connection_ok diagnosis device fault repair message to DGP <table><tr><th>OccupStatus</th></tr><tr><td>WAITING_ FOR_ SWEEPING_ TRAIN</td></tr><tr><td>SWEEPING_ TRAIN_ DETECTED</td></tr></table>				OccupStatus	WAITING_ FOR_ SWEEPING_ TRAIN	SWEEPING_ TRAIN_ DETECTED
OccupStatus						
WAITING_ FOR_ SWEEPING_ TRAIN						
SWEEPING_ TRAIN_ DETECTED						
Type:	FUR - Functional Requirement					
Status:	draft	Stability:	stable			
Target Release:	SEA SW GEN 3.0	Priority:	mandatory			
Originator:	Technics (R&D)	Allocation:	SW EC-MUX			
Verification Method:	Test	Verification Step:	module test, subit			
Comments:						
Satisfies:						

EC-MUX-AC:SSS:REQ: 2183	Axle Counter - Connection loss between EC-MUX and TDS after Reset command
Given an Axle Counter element with an open FC command with <FC_mode> mode When losing the connection between EC-MUX and TDS	

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Then SEA shall set the current element state to UNDEFINED
And SEA shall report the element state e_undefined_es with command reply flag true to CC
And SEA shall report the aspect diagnosis message AC_UNDEFINED to DGP
And SEA shall report the Axle Counter no_connection diagnosis device fault message to DGP

FC_ mode

FC_ C

FC_ U

FC_ P

Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-AC:SSS:REQ: 2182	Axle Counter - Reset command received from CC during EC-MUX-TDS disconnect
------------------------------------	---

Given the current element state is UNDEFINED
And the Axle Counter element is configured to support the <FC_mode> mode of the reset command
And CC connection state is CONNECTED
And EC-MUX - TDS connection state is NOT_CONNECTED

When receiving the command e_reset_rp from CC

Then SEA shall report the element state e_undefined_es with command reply flag true to CC
And SEA shall report the rejected command diagnosis to DGP

FC_ mode

FC_ C

FC_ U

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FC_mode FC_P			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

3.1.2.2 Diagnosis - Axle Counter

In order to provide the aspect diagnosis for the Axle Counter

As a SEA

I want to be able to receive an aspect diagnosis query from DGP and answer it by the DGP diagnosis values corresponding to the current element state of the queried Axle Counter

For requirement language see Chapter 2.6 "Requirement Language".

Given SEA with a configured Axle Counter element

And the mapping between SEA diagnosis aspect and DGP diagnosis value is defined

SEA diagnosis aspect	DGP diagnosis value
AC_FREE	0
AC_OCCUPIED_ABLE	1
AC_OCCUPIED_UNABLE	2
AC_OCCUPIED_TIMED	3
AC_DISTURBED_ABLE	4
AC_DISTURBED_UNABLE	5
AC_DISTURBED_TIMED	6
AC_UNDEFINED	7

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EC-MUX-AC:SSS:REQ: 2190	Axle Counter - aspect query																				
Given the current element state is <State> When receiving an aspect query from DGP Then SEA shall report the diagnosis aspect message <DGPMMessage> to DGP <table><tr><th>State</th><th>DGPMMessage</th></tr><tr><td>UNDEFINED</td><td>AC_ UNDEFINED</td></tr><tr><td>OCCUPIED_ ABLE</td><td>AC_ OCCUPIED_ ABLE</td></tr><tr><td>OCCUPIED_ UNABLE</td><td>AC_ OCCUPIED_ UNABLE</td></tr><tr><td>OCCUPIED_ TIMED</td><td>AC_ OCCUPIED_ TIMED</td></tr><tr><td>DISTURBED_ ABLE</td><td>AC_ DISTURBED_ ABLE</td></tr><tr><td>DISTURBED_ UNABLE</td><td>AC_ DISTURBED_ UNABLE</td></tr><tr><td>DISTURBED_ TIMED</td><td>AC_ DISTURBED_ TIMED</td></tr><tr><td>FREE</td><td>AC_ FREE</td></tr></table>				State	DGPMMessage	UNDEFINED	AC_ UNDEFINED	OCCUPIED_ ABLE	AC_ OCCUPIED_ ABLE	OCCUPIED_ UNABLE	AC_ OCCUPIED_ UNABLE	OCCUPIED_ TIMED	AC_ OCCUPIED_ TIMED	DISTURBED_ ABLE	AC_ DISTURBED_ ABLE	DISTURBED_ UNABLE	AC_ DISTURBED_ UNABLE	DISTURBED_ TIMED	AC_ DISTURBED_ TIMED	FREE	AC_ FREE
State	DGPMMessage																				
UNDEFINED	AC_ UNDEFINED																				
OCCUPIED_ ABLE	AC_ OCCUPIED_ ABLE																				
OCCUPIED_ UNABLE	AC_ OCCUPIED_ UNABLE																				
OCCUPIED_ TIMED	AC_ OCCUPIED_ TIMED																				
DISTURBED_ ABLE	AC_ DISTURBED_ ABLE																				
DISTURBED_ UNABLE	AC_ DISTURBED_ UNABLE																				
DISTURBED_ TIMED	AC_ DISTURBED_ TIMED																				
FREE	AC_ FREE																				
Type:	FUR - Functional Requirement																				
Status:	draft	Stability:	stable																		
Target Release:	SEA SW GEN 3.0	Priority:	mandatory																		
Originator:	Technics (R&D)	Allocation:	SW EC-MUX																		
Verification Method:	Test	Verification Step:	module test, subit																		
Comments:																					
Satisfies:																					

3.1.2.3 System - Axle Counter

In order to provide the internal state of an Axle Counter
As a SEA
I want to be able to serialize/deserialize element state when a state transfer during EC-MUX redundancy integration takes place.

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Given SEA with a configured Axle Counter element

EC-MUX-AC:SSS:REQ:2196	Axle Counter - state transfer		
When a state transfer during EC-MUX redundancy integration takes place			
Then the element process shall serialize/deserialize element state considering: internal element state			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test, Inspection/Review	Verification Step:	module test, code reading
Comments:	For the Axle Counter, during an EC-MUX redundancy integration when the state transfer takes place the element process has to be capable to serialize a state and deserialize from a state with the information of the current internal element state.		
Satisfies:	SEA:SRS:REQ:0813, SEA:SRS:REQ:0814		

3.2 Requirements Traceability

Coverage of the SEA system-level requirements [6] by the SEA SW Subsystem Train Detection System is provided in a separate document: see Verification Cross Reference Matrix [2]

TODO: Reference to be updated.

3.2.1 Requirements List

Identifier	Synopsis	Category	Status
EC-MUX-AC:SSS:REQ:2196	Axle Counter - state transfer	FUR - Functional Requirement	draft
EC-MUX-AC:SSS:REQ:2136	Axle Counter - initial state	FUR - Functional Requirement	draft
EC-MUX-AC:SSS:	Axle Counter - Spontaneous message - State transition with	FUR -	

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Identifier	Synopsis	Category	Status
REQ:2137	reporting without fault information from FREE	Functional Requirement	draft
EC-MUX-AC:SSS: REQ:2188	Axle Counter - Spontaneous message - Ignore repeated VACANT message in FREE	FUR - Functional Requirement	draft
EC-MUX-AC:SSS: REQ:2187	Axle Counter - Spontaneous message - State transition with reporting without fault information from UNDEFINED	FUR - Functional Requirement	draft
EC-MUX-AC:SSS: REQ:2138	Axle Counter - Spontaneous message - State transition with reporting with fault information	FUR - Functional Requirement	draft
EC-MUX-AC:SSS: REQ:2186	Axle Counter - Spontaneous message - State transition with reporting with fault information from UNDEFINED	FUR - Functional Requirement	draft
EC-MUX-AC:SSS: REQ:2161	Axle Counter - Spontaneous message - Ignore repeated sweeping train messages in UNDEFINED	FUR - Functional Requirement	draft
EC-MUX-AC:SSS: REQ:2139	Axle Counter - Spontaneous message - State transition without reporting and without fault information	FUR - Functional Requirement	draft
EC-MUX-AC:SSS: REQ:2160	Axle Counter - Spontaneous message - Ignore repeated messages	FUR - Functional Requirement	draft
EC-MUX-AC:SSS: REQ:2140	Axle Counter - Spontaneous message - State transition to FREE with timer	FUR - Functional Requirement	draft
EC-MUX-AC:SSS: REQ:2141	Axle Counter - Spontaneous message - State transition to FREE with timer and fault information	FUR - Functional Requirement	draft
EC-MUX-AC:SSS: REQ:2142	Axle Counter - Spontaneous message - State transition with timer cancelling and fault information	FUR - Functional Requirement	draft
EC-MUX-AC:SSS: REQ:2143	Axle Counter - Spontaneous message - State transition with timer cancelling and without fault information	FUR - Functional Requirement	draft
EC-MUX-AC:SSS: REQ:2154	Axle Counter - Spontaneous message - Ignore Occupancy Status VACANT in timed state	FUR - Functional Requirement	draft
EC-MUX-AC:SSS: REQ:2144	Axle Counter - Command reply - State transition after successful FC command with reporting and without fault information	FUR - Functional Requirement	draft
EC-MUX-AC:SSS: REQ:2145	Axle Counter - Command reply - State transition after successful FC command with reporting and with fault information	FUR - Functional Requirement	draft
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Identifier	Synopsis	Category	Status
EC-MUX-AC:SSS:REQ:2146	Axle Counter - Command reply - State transition with reporting without fault information	Functional Requirement	draft
EC-MUX-AC:SSS:REQ:2147	Axle Counter - Command reply - State transition with reporting with fault occurrence information	FUR - Functional Requirement	draft
EC-MUX-AC:SSS:REQ:2195	Axle Counter - Command reply - State transition with reporting with fault repair information	FUR - Functional Requirement	draft
EC-MUX-AC:SSS:REQ:2148	Axle Counter - Command reply - Transition by sweeping train state	FUR - Functional Requirement	draft
EC-MUX-AC:SSS:REQ:2149	Axle Counter - Command reply - Close open FC_P command with sweeping train state	FUR - Functional Requirement	draft
EC-MUX-AC:SSS:REQ:2194	Axle Counter - Command Reply - Close open FC_P command with repeated OCCUPIED / DISTURBED state	FUR - Functional Requirement	draft
EC-MUX-AC:SSS:REQ:2155	Axle Counter - Command reply - Ignore sweeping train as reply to FC_C / FC_U	FUR - Functional Requirement	draft
EC-MUX-AC:SSS:REQ:2184	Axle Counter - Spontaneous message - Ignore sweeping train status message in UNDEFINED state	FUR - Functional Requirement	draft
EC-MUX-AC:SSS:REQ:2185	Axle Counter - Spontaneous message - Ignore sweeping train state with no open command	FUR - Functional Requirement	draft
EC-MUX-AC:SSS:REQ:2157	Axle Counter - Spontaneous message - Ignore message Additional Information	FUR - Functional Requirement	draft
EC-MUX-AC:SSS:REQ:2158	Axle Counter - Spontaneous message - Disconnect on message TDP Status	FUR - Functional Requirement	draft
EC-MUX-AC:SSS:REQ:2190	Axle Counter - aspect query	FUR - Functional Requirement	draft
EC-MUX-AC:SSS:REQ:2174	Axle Counter - [Re-]Connecting to CC - TDS CONNECTED - General inquiry	FUR - Functional Requirement	draft
EC-MUX-AC:SSS:REQ:2175	Axle Counter - [Re-]Connecting to CC - TDS NOT_CONNECTED - General inquiry	FUR - Functional Requirement	draft
EC-MUX-AC:SSS:REQ:2176	Axle Counter - Connection loss to CC	FUR - Functional Requirement	draft
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Identifier	Synopsis	Category	Status
EC-MUX-AC:SSS:REQ:2179	Axle Counter - Connection loss between EC-MUX and TDS in regular states	Functional Requirement	draft
EC-MUX-AC:SSS:REQ:2180	Axle Counter - Connection loss between EC-MUX and TDS in disturbed states	FUR - Functional Requirement	draft
EC-MUX-AC:SSS:REQ:2181	Axle Counter - Connection loss between EC-MUX and TDS - CC not connected	FUR - Functional Requirement	draft
EC-MUX-AC:SSS:REQ:2177	Axle Counter - Connection loss between EC-MUX and TDS - UNDEFINED	FUR - Functional Requirement	draft
EC-MUX-AC:SSS:REQ:2178	Axle Counter - [Re-]Connecting between EC-MUX and TDS - TDS in VACANT / OCCUPIED state	FUR - Functional Requirement	draft
EC-MUX-AC:SSS:REQ:2198	Axle Counter - [Re-]Connecting between EC-MUX and TDS - TDS in DISTURBED state	FUR - Functional Requirement	draft
EC-MUX-AC:SSS:REQ:2197	Axle Counter - [Re-]Connecting between EC-MUX and TDS - TDS in sweeping train state	FUR - Functional Requirement	draft
EC-MUX-AC:SSS:REQ:2183	Axle Counter - Connection loss between EC-MUX and TDS after Reset command	FUR - Functional Requirement	draft
EC-MUX-AC:SSS:REQ:2182	Axle Counter - Reset command received from CC during EC-MUX-TDS disconnect	FUR - Functional Requirement	draft
EC-MUX-AC:SSS:REQ:2156	Axle Counter - Consistent Disturbance Status without error reporting in disturbed states	FUR - Functional Requirement	draft
EC-MUX-AC:SSS:REQ:2162	Axle Counter - Consistent Disturbance Status without error reporting in regular states	FUR - Functional Requirement	draft
EC-MUX-AC:SSS:REQ:2163	Axle Counter - Disconnect on Inconsistent Disturbance Status in disturbed states	FUR - Functional Requirement	draft
EC-MUX-AC:SSS:REQ:2164	Axle Counter - Disconnect on Inconsistent Disturbance Status in regular states	FUR - Functional Requirement	draft
EC-MUX-AC:SSS:REQ:2165	Axle Counter - Inconsistent POM Status with error reporting	FUR - Functional Requirement	draft
EC-MUX-AC:SSS:REQ:2166	Axle Counter - Consistent POM Status without error reporting	FUR - Functional Requirement	draft
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Identifier	Synopsis	Category	Status
EC-MUX-AC:SSS:REQ:2167	Axle Counter - Inconsistent Filling Level with error reporting	Functional Requirement	draft
EC-MUX-AC:SSS:REQ:2168	Axle Counter - Consistent Filling Level without error reporting	FUR - Functional Requirement	draft
EC-MUX-AC:SSS:REQ:2169	Axle Counter - Consistent Change Trigger without error reporting	FUR - Functional Requirement	draft
EC-MUX-AC:SSS:REQ:2170	Axle Counter - Disconnect on Inconsistent Change Trigger in regular states	FUR - Functional Requirement	draft
EC-MUX-AC:SSS:REQ:2171	Axle Counter - Consistent Change Trigger without error reporting in disturbed states	FUR - Functional Requirement	draft
EC-MUX-AC:SSS:REQ:2172	Axle Counter - Inconsistent Ability in VACANT message with error reporting	FUR - Functional Requirement	draft
EC-MUX-AC:SSS:REQ:2191	Axle Counter - Consistent Ability in VACANT message	FUR - Functional Requirement	draft
EC-MUX-AC:SSS:REQ:2173	Axle Counter - Inconsistent Ability in sweeping train message with error reporting	FUR - Functional Requirement	draft
EC-MUX-AC:SSS:REQ:2192	Axle Counter - Consistent Ability in sweeping train message	FUR - Functional Requirement	draft
EC-MUX-AC:SSS:REQ:2193	Axle Counter - Consistent sweeping train status message with any Disturbance Status	FUR - Functional Requirement	draft
EC-MUX-AC:SSS:REQ:2130	Axle Counter - Execute reset command	FUR - Functional Requirement	draft
EC-MUX-AC:SSS:REQ:2131	Axle Counter - Reject reset command	FUR - Functional Requirement	draft
EC-MUX-AC:SSS:REQ:2132	Axle Counter - command reply with message Command Rejected from TDS	FUR - Functional Requirement	draft
EC-MUX-AC:SSS:REQ:2133	Axle Counter - command reply with message TVPS FC_P Failed	FUR - Functional Requirement	draft
EC-MUX-AC:SSS:REQ:2134	Axle Counter - Spontaneous message - Disconnect on Command Rejected in UNDEFINED	FUR - Functional Requirement	draft
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Identifier	Synopsis	Category	Status
EC-MUX-AC:SSS:REQ:2135	Axle Counter - Spontaneous message - do nothing on Command Rejected	Functional Requirement	draft
EC-MUX-AC:SSS:REQ:2150	Axle Counter - reject repeated command	FUR - Functional Requirement	draft
EC-MUX-AC:SSS:REQ:2151	Axle Counter - Command Reply - Disconnect on TVPS FC_P_A Failed message	FUR - Functional Requirement	draft
EC-MUX-AC:SSS:REQ:2159	Axle Counter - Spontaneous message - Disconnect on TVPS FC_P_A Failed message	FUR - Functional Requirement	draft
EC-MUX-AC:SSS:REQ:2152	Axle Counter - Command Reply - Disconnect on TVPS FC_P Failed message	FUR - Functional Requirement	draft
EC-MUX-AC:SSS:REQ:2153	Axle Counter - Spontaneous Message - Disconnect on TVPS FC_P Failed message	FUR - Functional Requirement	draft

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